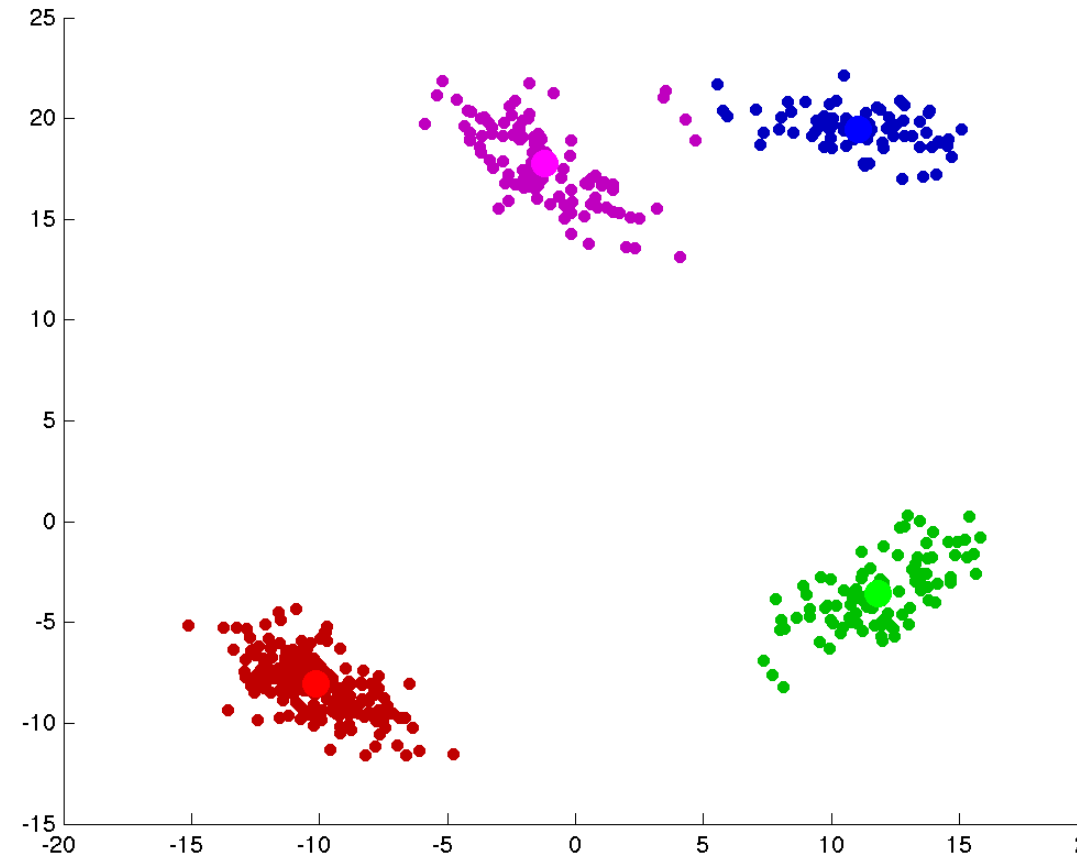


CPSC 340: Machine Learning and Data Mining

More Clustering

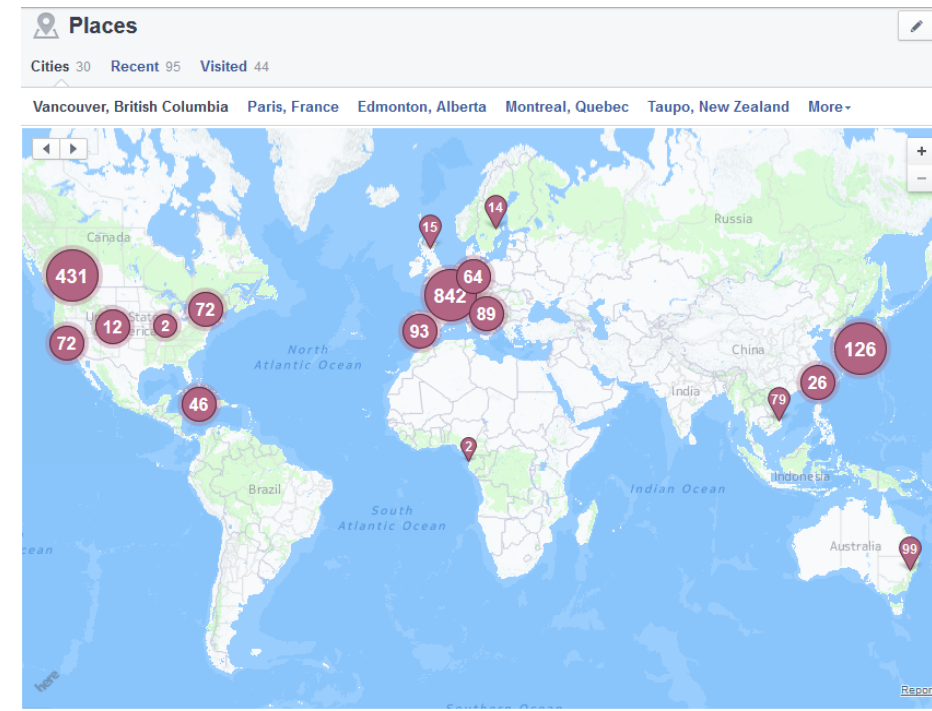
Last Time: K-Means Clustering

- We want to **cluster** data:
 - Assign examples to groups.
- **K-means clustering**:
 - Define groups by “means”
 - Assigns examples to nearest mean.
(And updates means during training.)
- Issues with k-means:
 - Fast but sensitive to initialization.
 - Choosing ‘k’ is annoying.



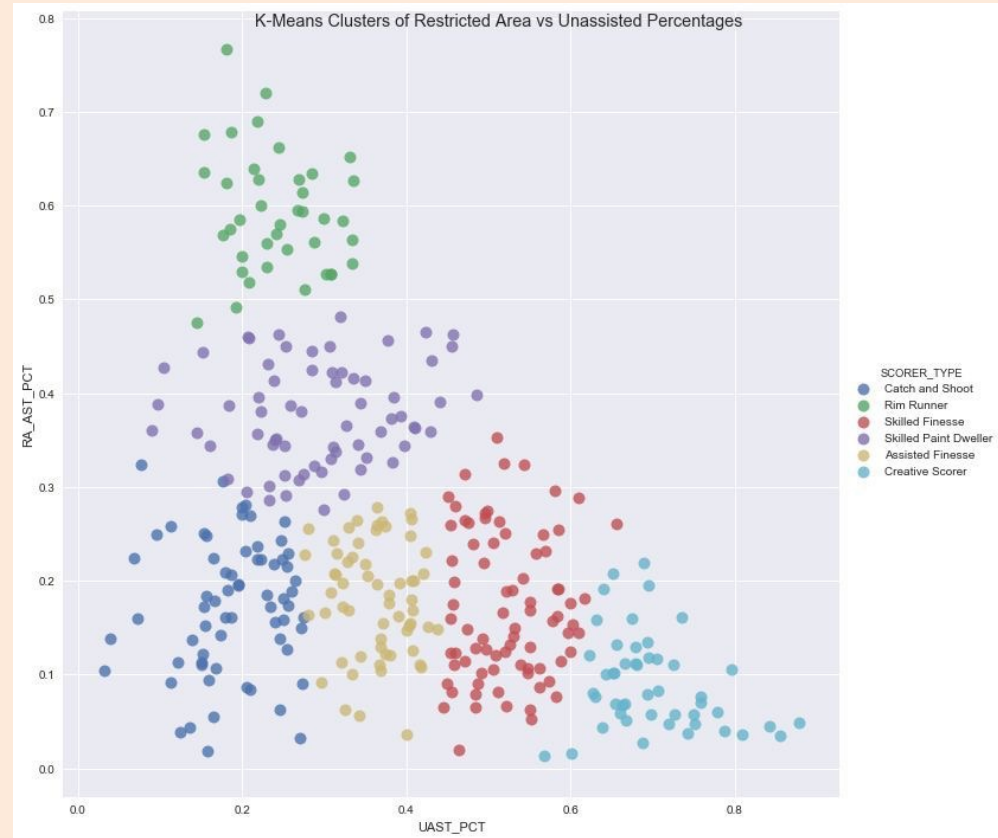
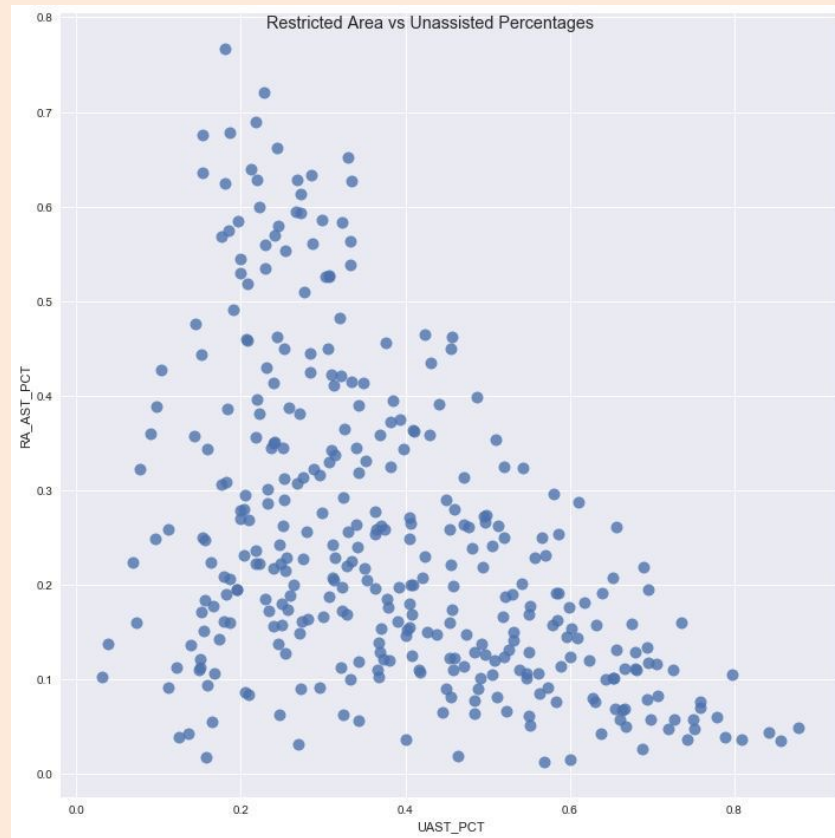
Vector Quantization

- K-means originally comes from signal processing.
- Designed for **vector quantization**:
 - Replace examples with the mean of their cluster (“prototype”).
- Example:
 - Facebook places: 1 location summarizes many.
 - What sizes of clothing should I make?



Vector Quantization for Basketball Players

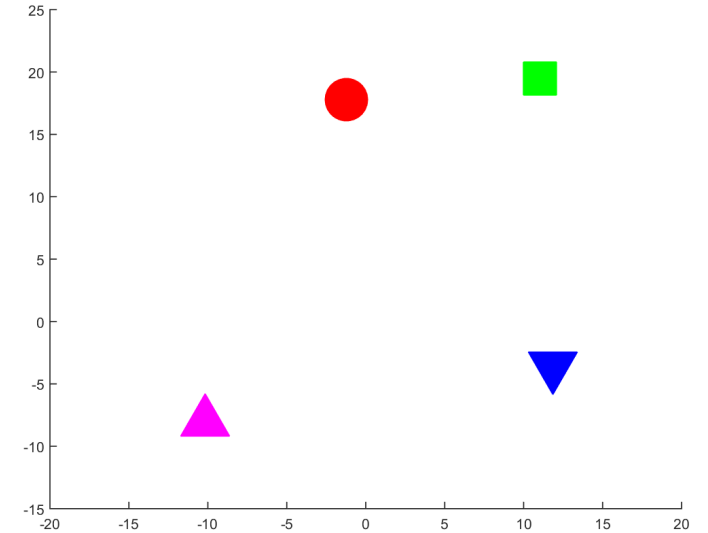
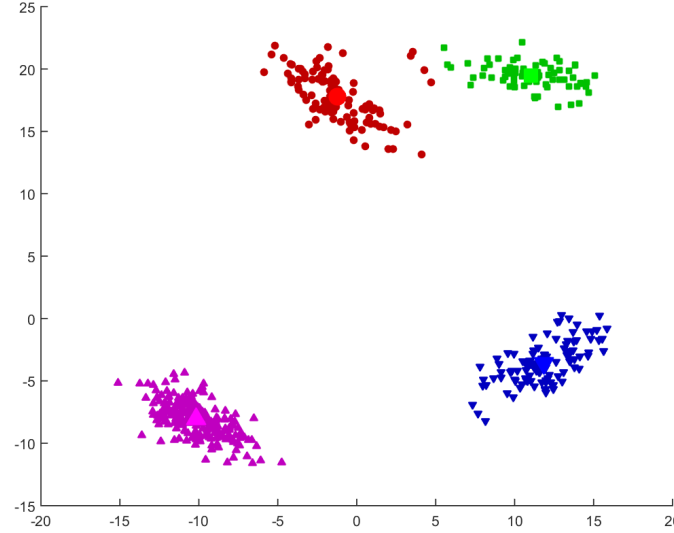
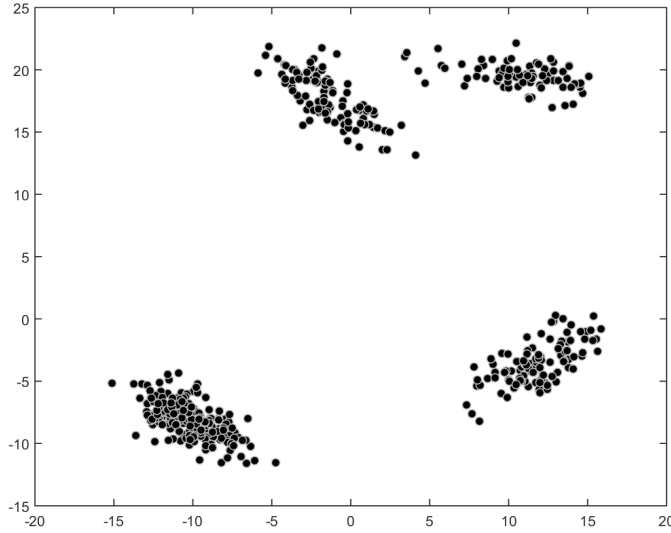
- Clustering NBA basketball players based on shot type/percentage:



- The “prototypes” (means) give offensive styles (like “catch and shoot”).

The name does not make sense by OK!

Vector Quantization Example



$n \times d$

$$X = \begin{bmatrix} -9.0 & -7.3 \\ -10.9 & -9.0 \\ 13.7 & 19.3 \\ 13.8 & 20.4 \\ 12.8 & 20.6 \\ \vdots & \vdots \end{bmatrix}$$

Run k-means

$k \times d$

$$W = \begin{bmatrix} -1.2 & 17.8 \\ -10.2 & -8.0 \\ 11.0 & 19.5 \\ 11.8 & -3.6 \end{bmatrix}$$

$n \times 1$

$$\hat{y} = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 3 \\ 3 \\ \vdots \end{bmatrix}$$

Approximate objects with means.

$$\hat{X} = \begin{bmatrix} -10.2 & -8.0 \\ -10.2 & -8.0 \\ 11.0 & 19.5 \\ 11.0 & 19.5 \\ -1.2 & 17.8 \\ \vdots & \vdots \end{bmatrix}$$

these mean values are the prototypes. Assume each example can be approximated by prototype

(Bad) Vector Quantization in Practice

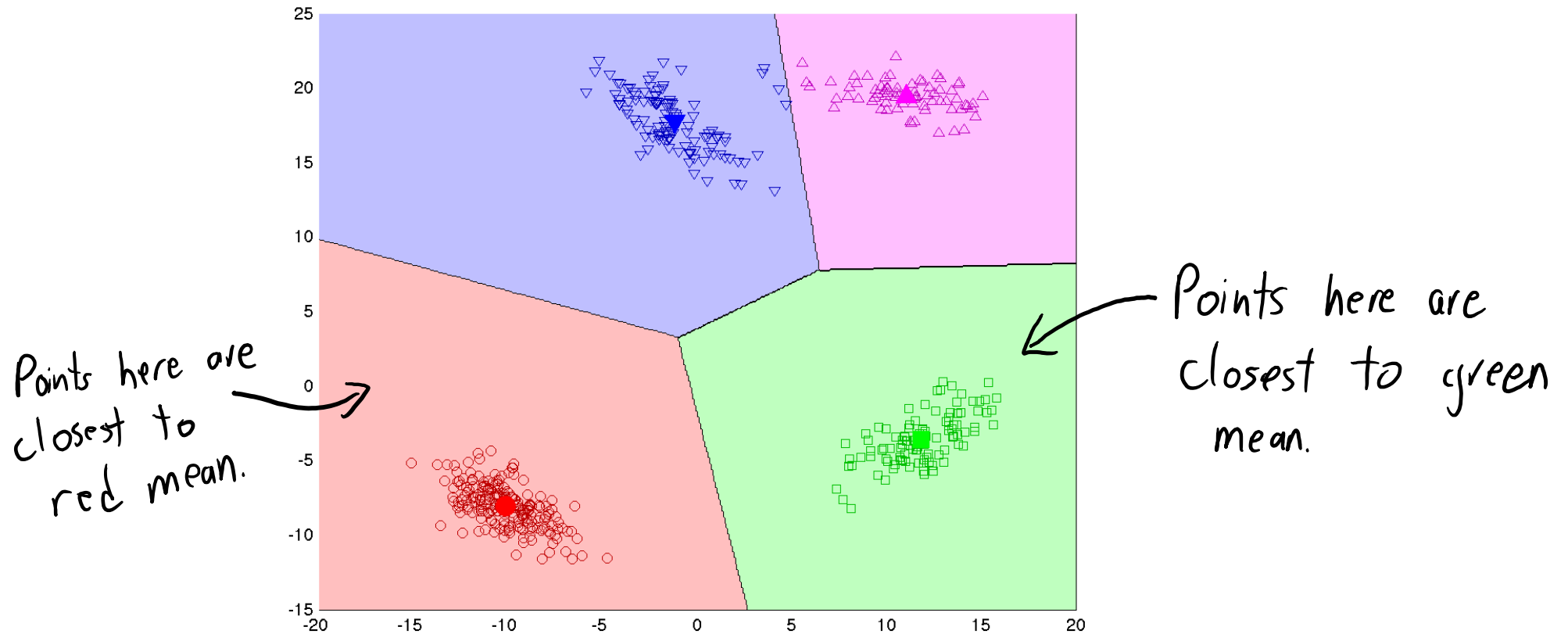
- Political parties can be thought as a form of vector quantization:



- Hope is that parties represent what a cluster of voters want.
 - With larger 'k' more voters have a party that closely reflects them.
 - With smaller 'k', parties are less accurate reflections of people's views.

Shape of K-Means Clusters

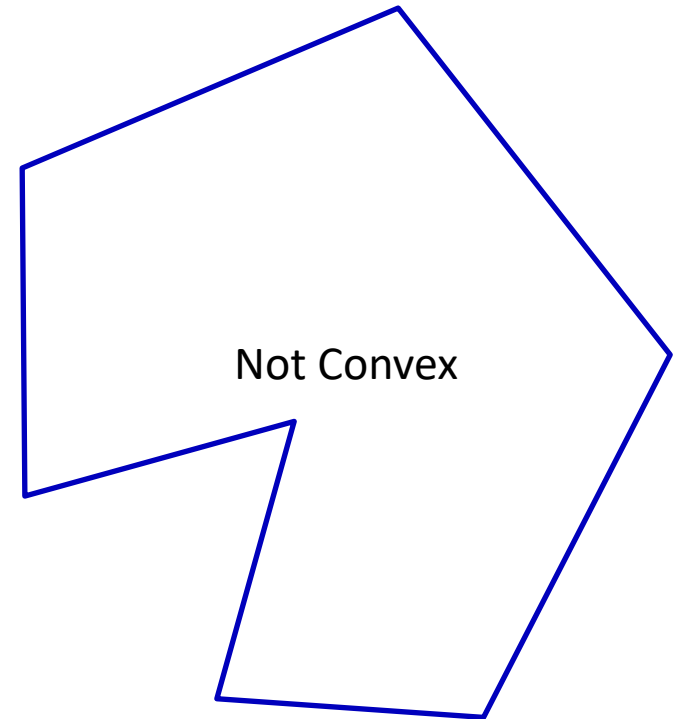
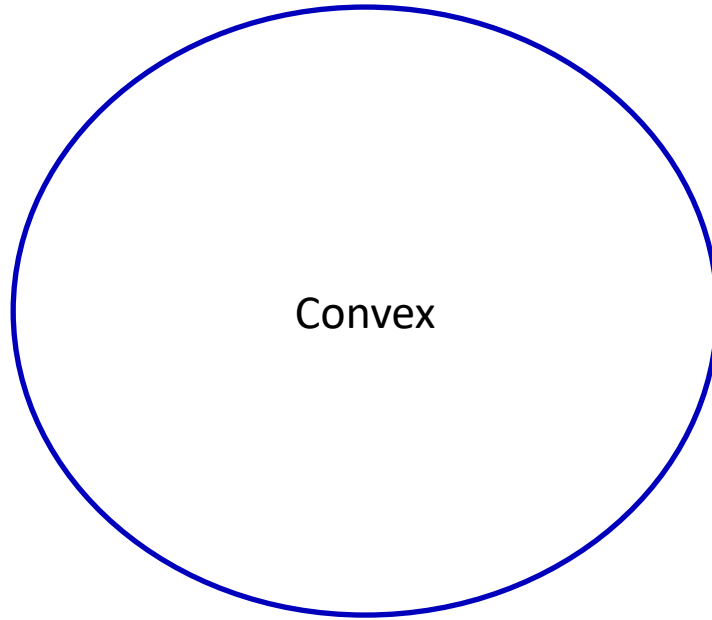
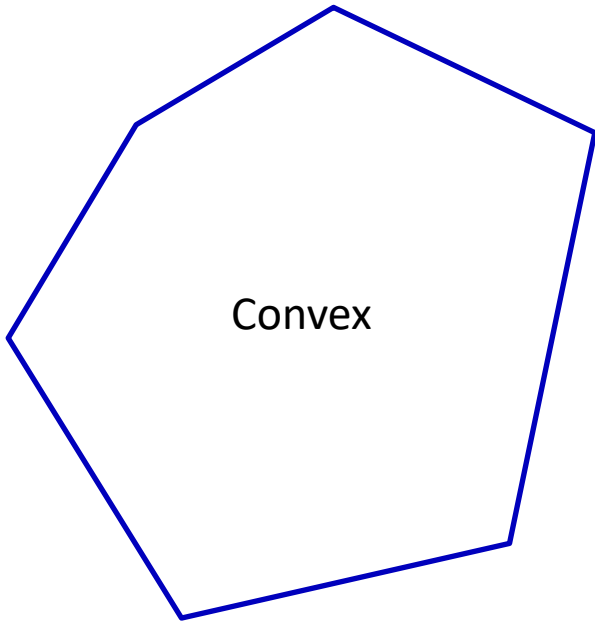
- Recall that k-means assigns cluster based on nearest mean.
- This leads to **partition the space** :



- Observe that the **clusters are convex** regions (proof in bonus).

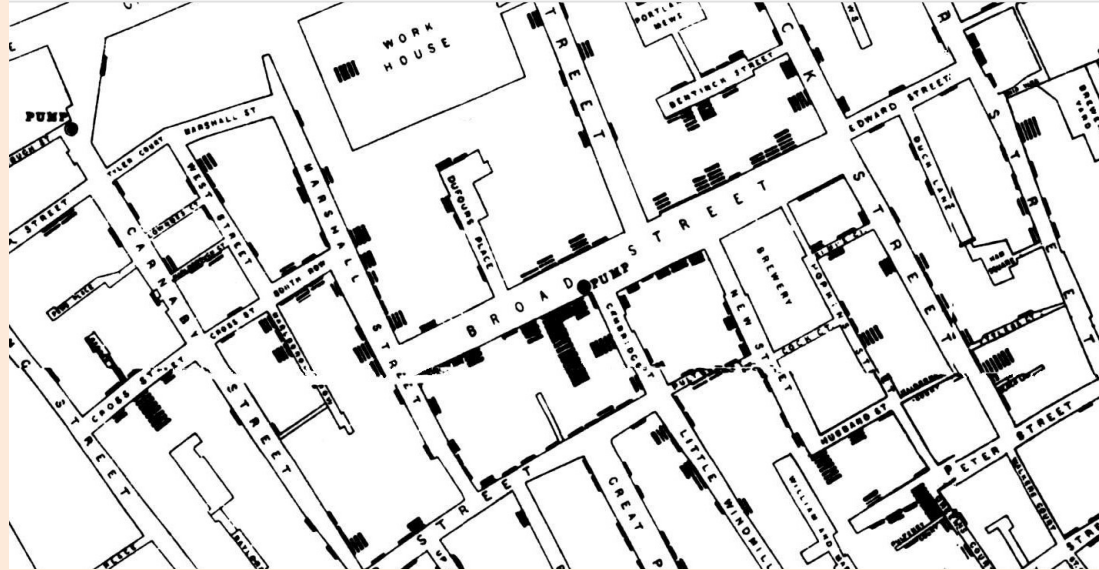
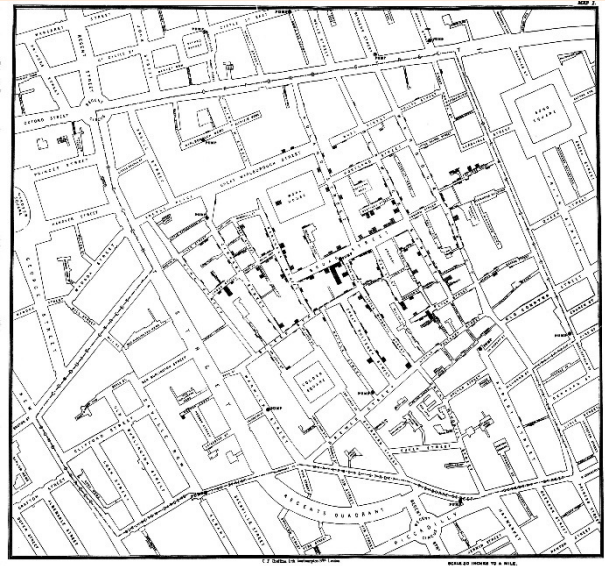
Convex Sets

- A set is **convex** if **line between two points in the set stays in the set**.



John Snow and Cholera Epidemic

- John Snow's 1854 spatial histogram of deaths from cholera:



- Found cluster of cholera deaths around a particular water pump.
 - Went against airborne theory, but pump later found to be contaminated.
 - “Father” of epidemiology.

john snow - Google Sea

← → ↺ 🏠 🔒 google.ca/search?safe=off&tbm=isch&q=john+snow&nfr=1&sa=X&ved=0ahUKEwj9gsnWsLXWAhUJ8mMKHUAQ8dQQvgUIkAloAQ&biw=1440&bih=750&d

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Did you mean: [jon snow](#)

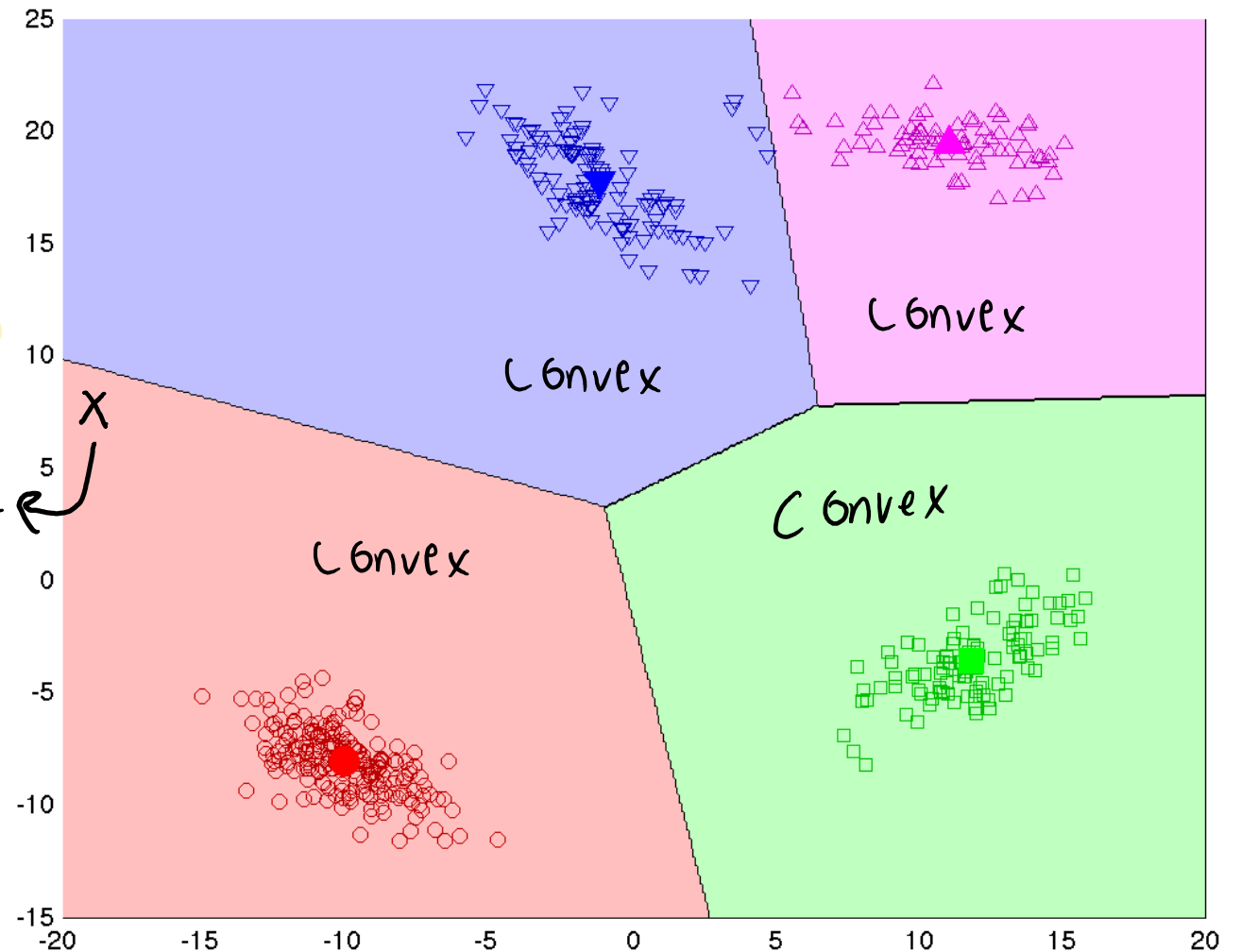


Shape of K-Means Clusters

Issues with shape of K-means clusters:

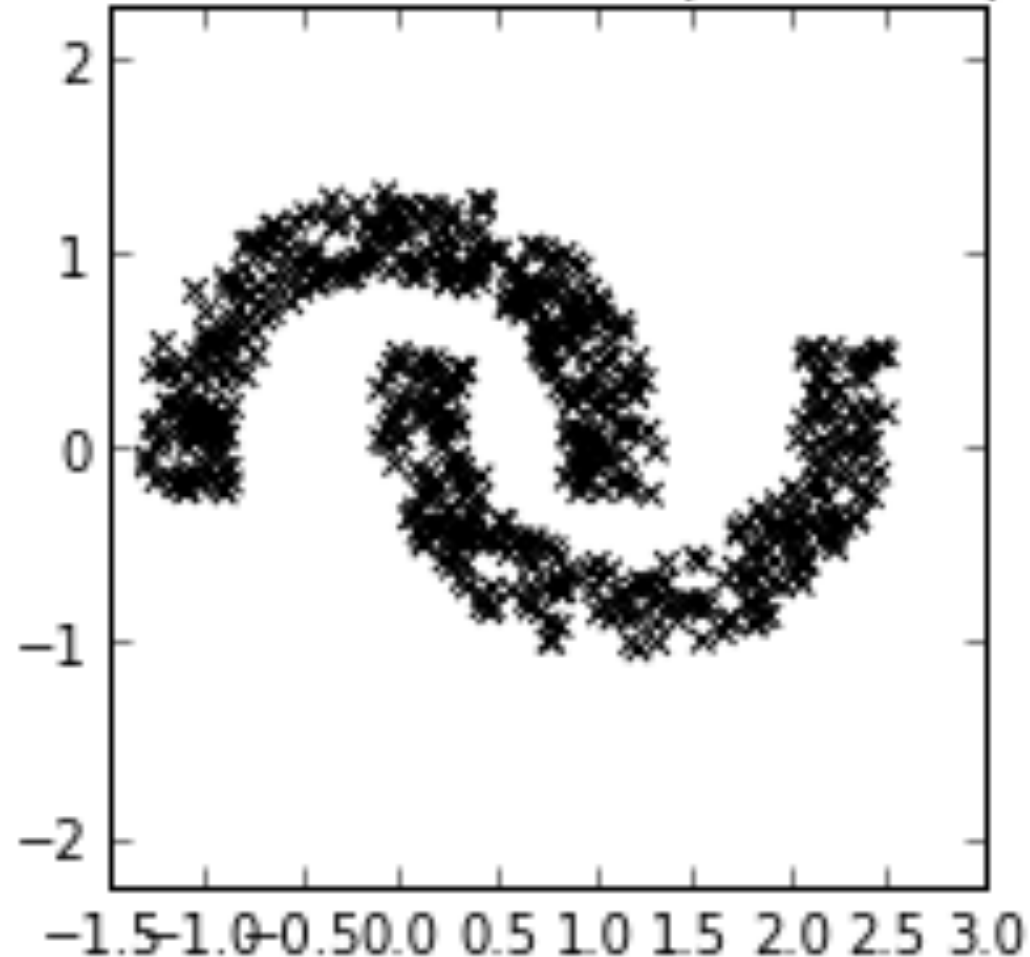
1. Clusters in the data might not be convex.

2. Does this point really belong in red cluster?

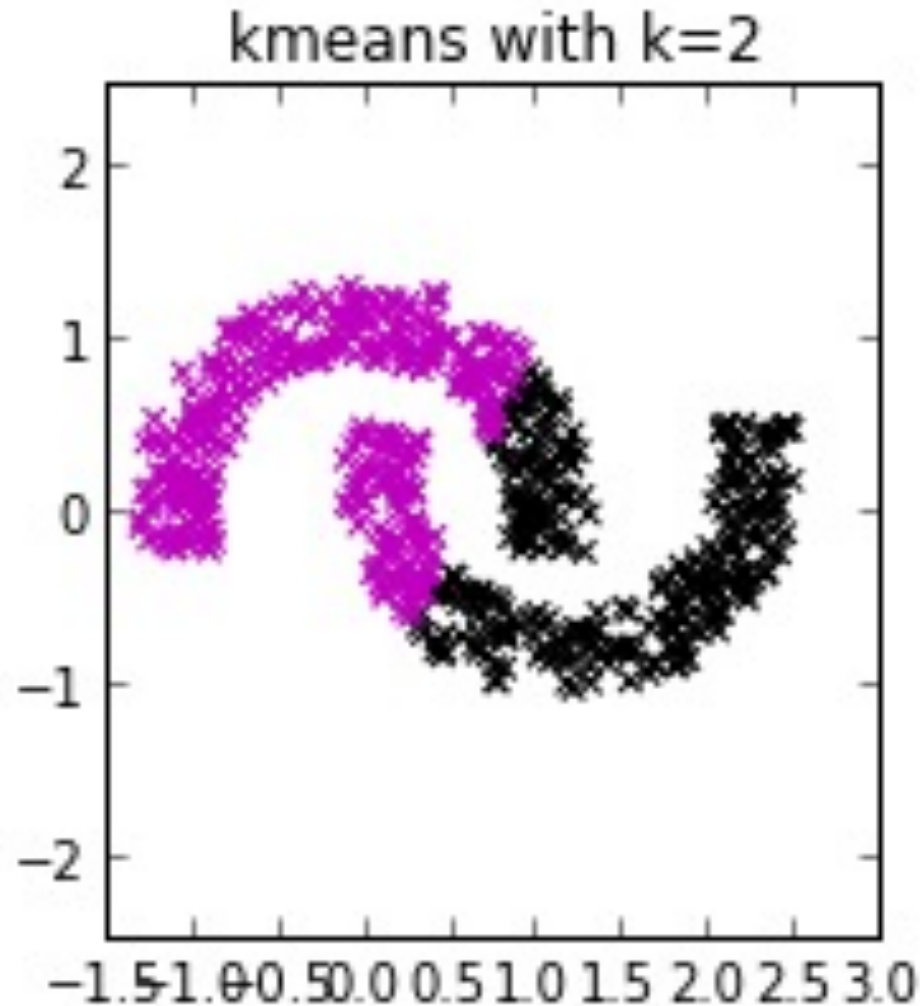


K-Means with Non-Convex Clusters

Non-convex banana-shaped data points

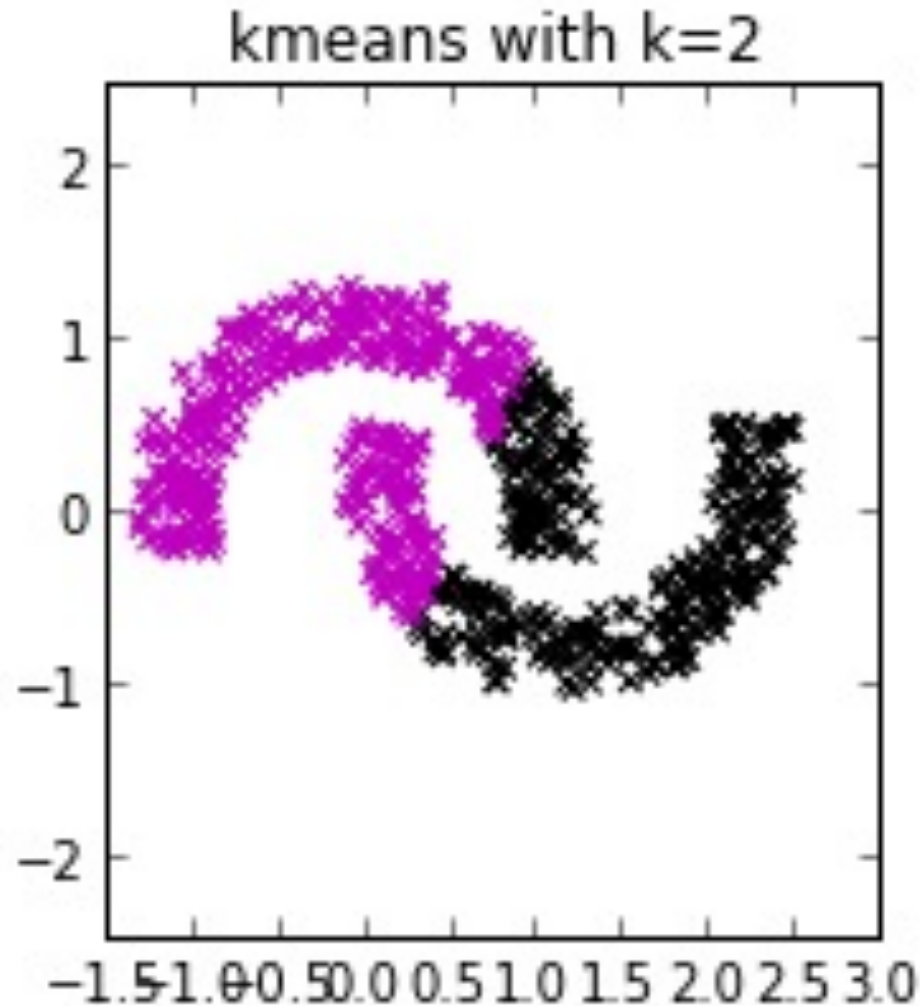


K-Means with Non-Convex Clusters



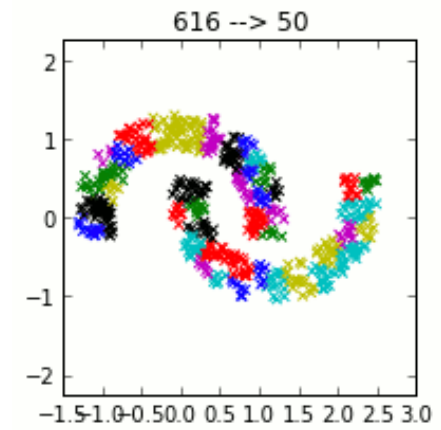
K-means **cannot separate**
some non-convex clusters

K-Means with Non-Convex Clusters



K-means **cannot separate**
some non-convex clusters

Though over-clustering can help
("hierarchical")



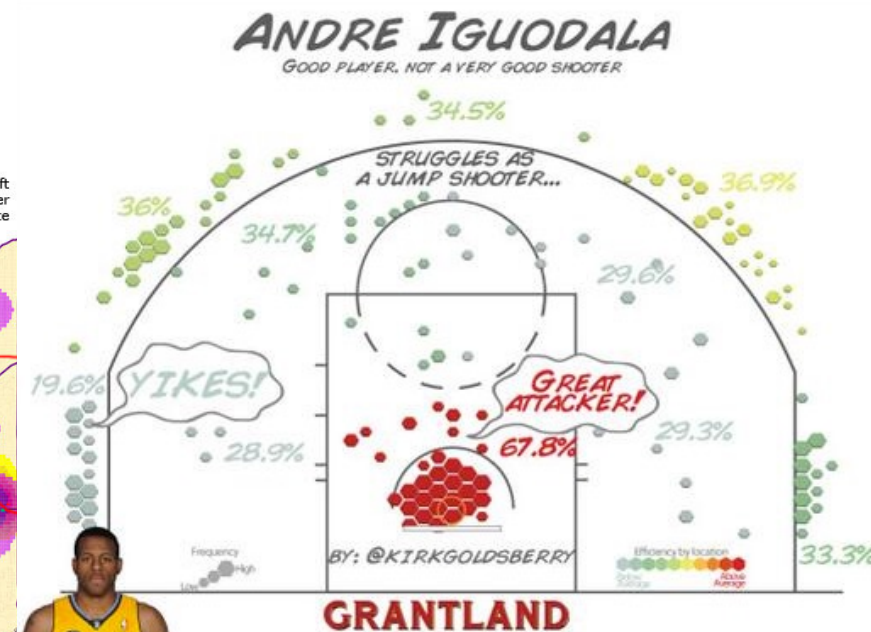
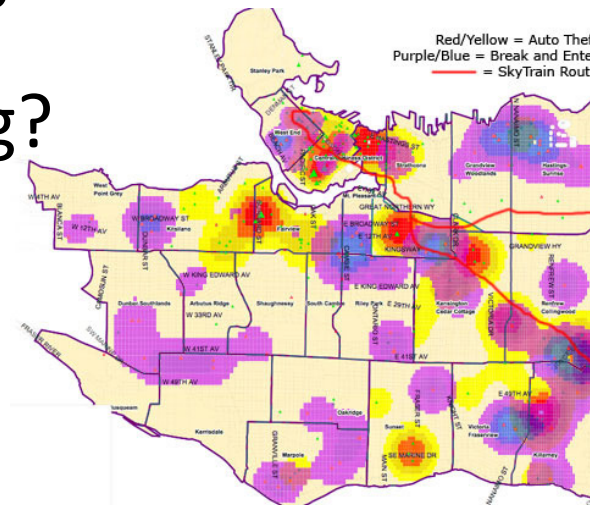
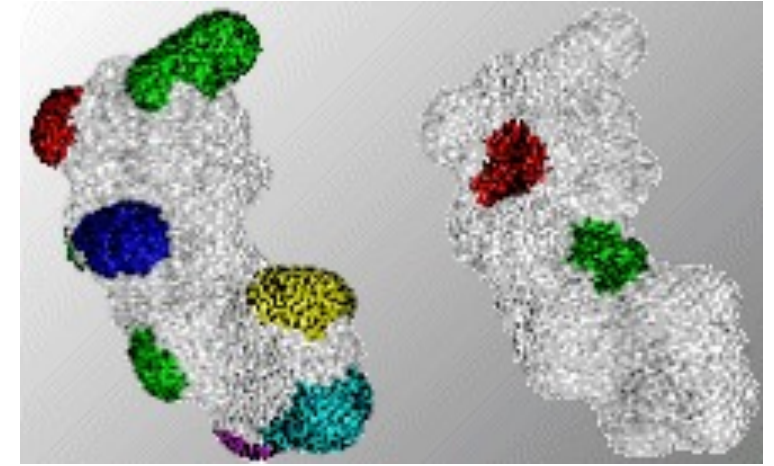
Motivation for Density-Based Clustering

- Density-based clustering:
 - Clusters are defined by “dense” regions.
 - Examples in non-dense regions don’t get clustered.
 - Not trying to “partition” the space.
- Clusters can be non-convex:
 - Elephant clusters affected by vegetation, mountains, rivers, water access, etc.
- It’s a non-parametric clustering method:
 - No fixed number of clusters ‘k’.
 - Clusters can become more complicated with more data.

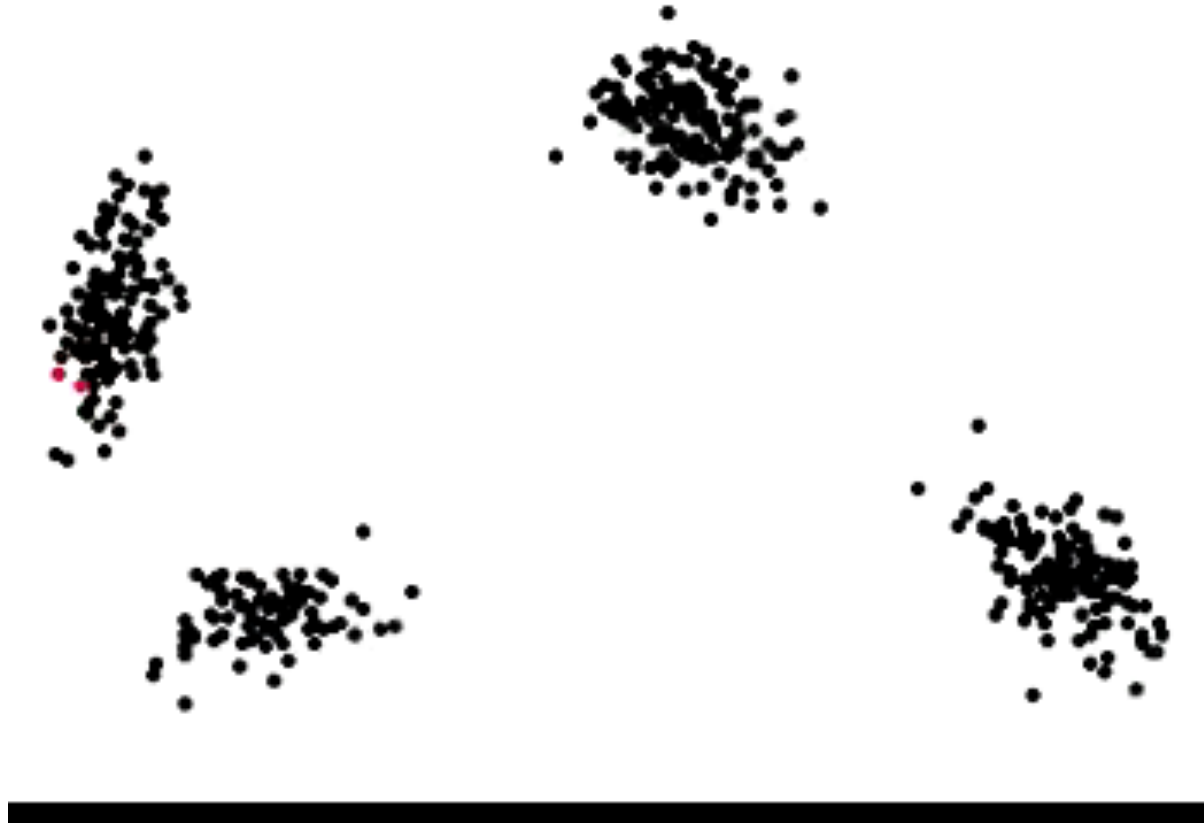


Other Potential Applications

- Where are high crime regions of a city?
- Where should taxis patrol?
- Where does Iguodala make/miss shots?
- Which products are similar to this one?
- Which pictures are in the same place?
- Where can proteins 'dock'?
- Where are people tweeting?



Density-Based Clustering in Action

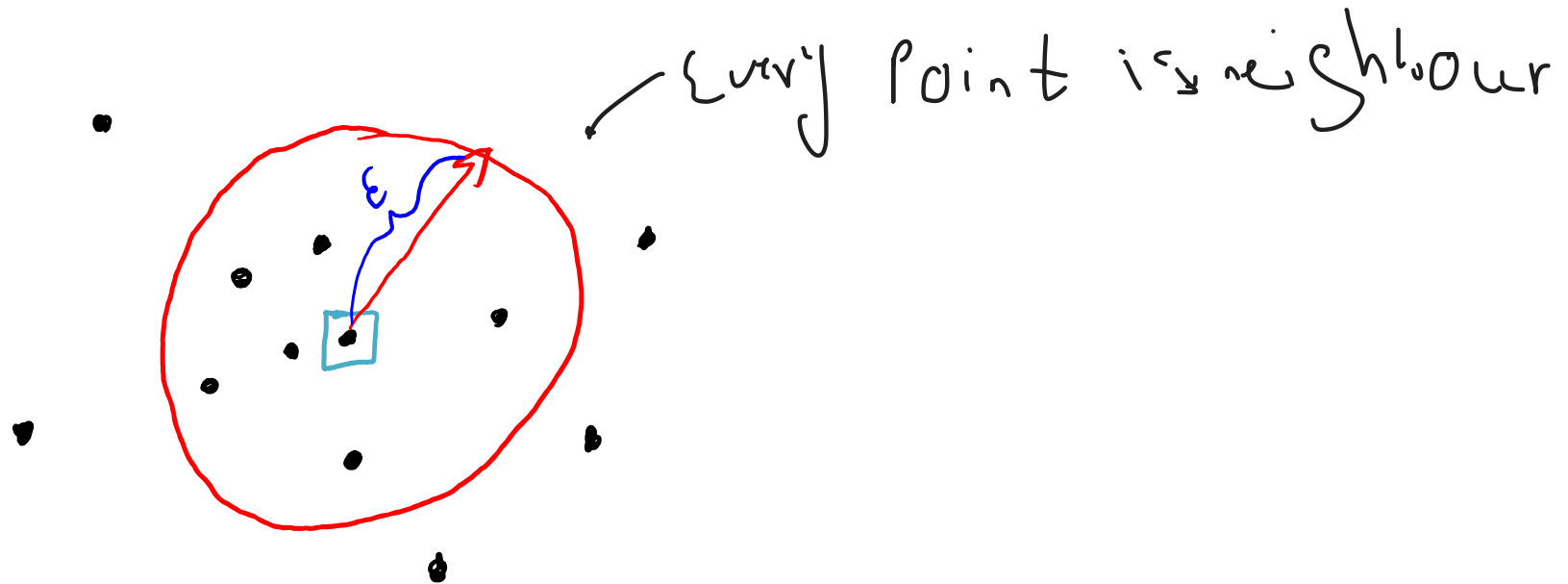


Density-Based Clustering

DBSCAN: "Density-Based Spatial
Clustering of Applications with Noise."

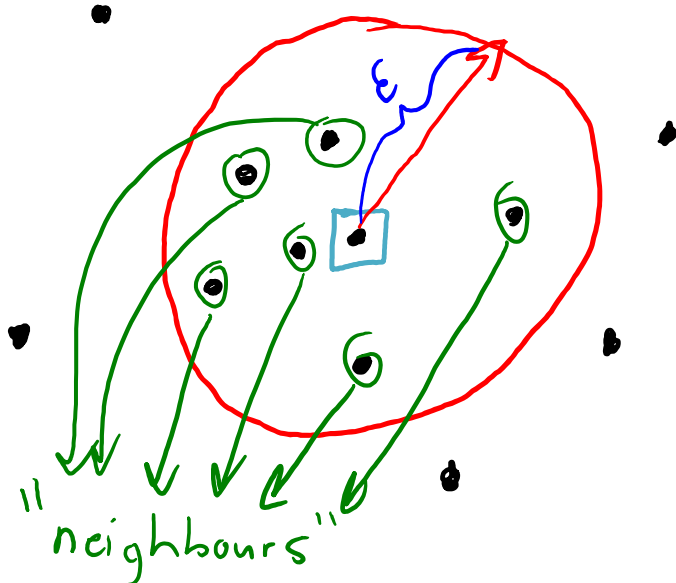
- Density-based clustering algorithm (DBSCAN) has two hyperparameters:
 - Epsilon (ϵ): distance we use to decide if another point is a "neighbour".

It is non-Parametrics



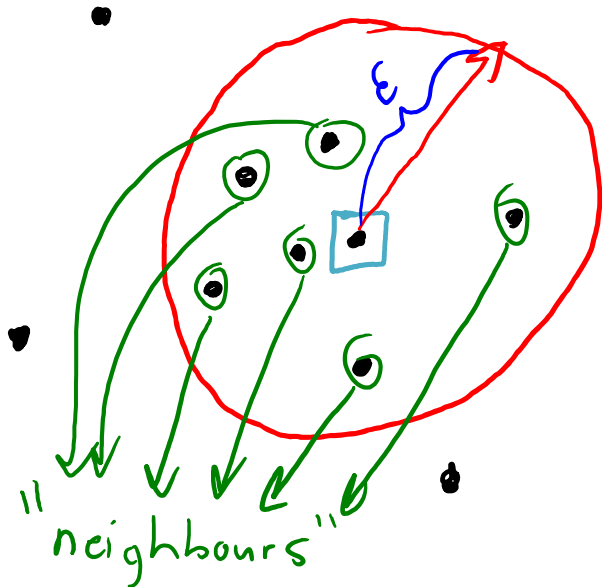
Density-Based Clustering

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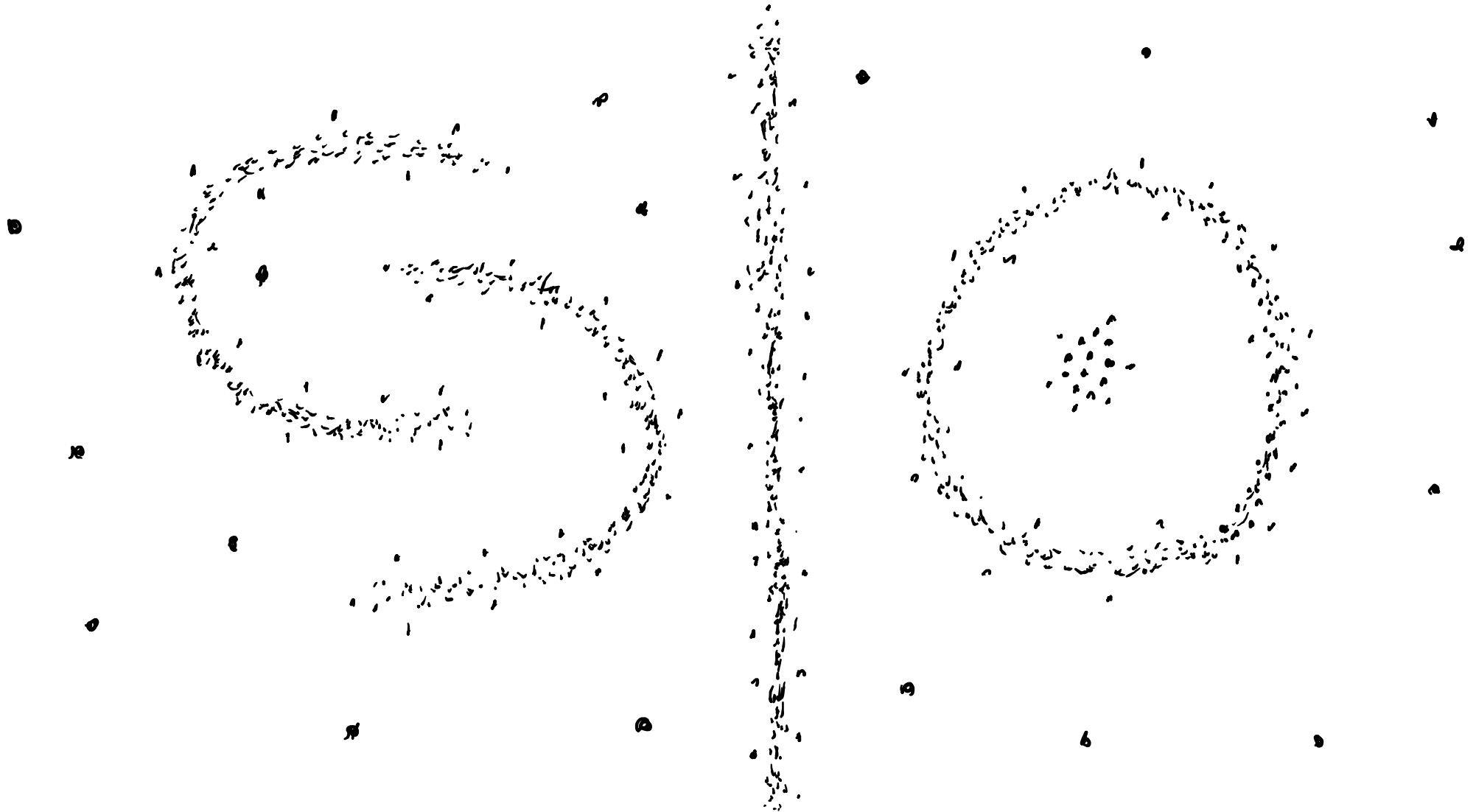
Density-Based Clustering

- Density-based clustering algorithm (DBSCAN) has two hyperparameters:
 - Epsilon (ϵ): distance we use to decide if another point is a “neighbour”.
 - MinNeighbours: number of neighbours needed to say a region is “dense”.
 - If you have at least minNeighbours “neighbours”, you are called a “core” point.
- Main idea: merge all neighbouring core points to form clusters.

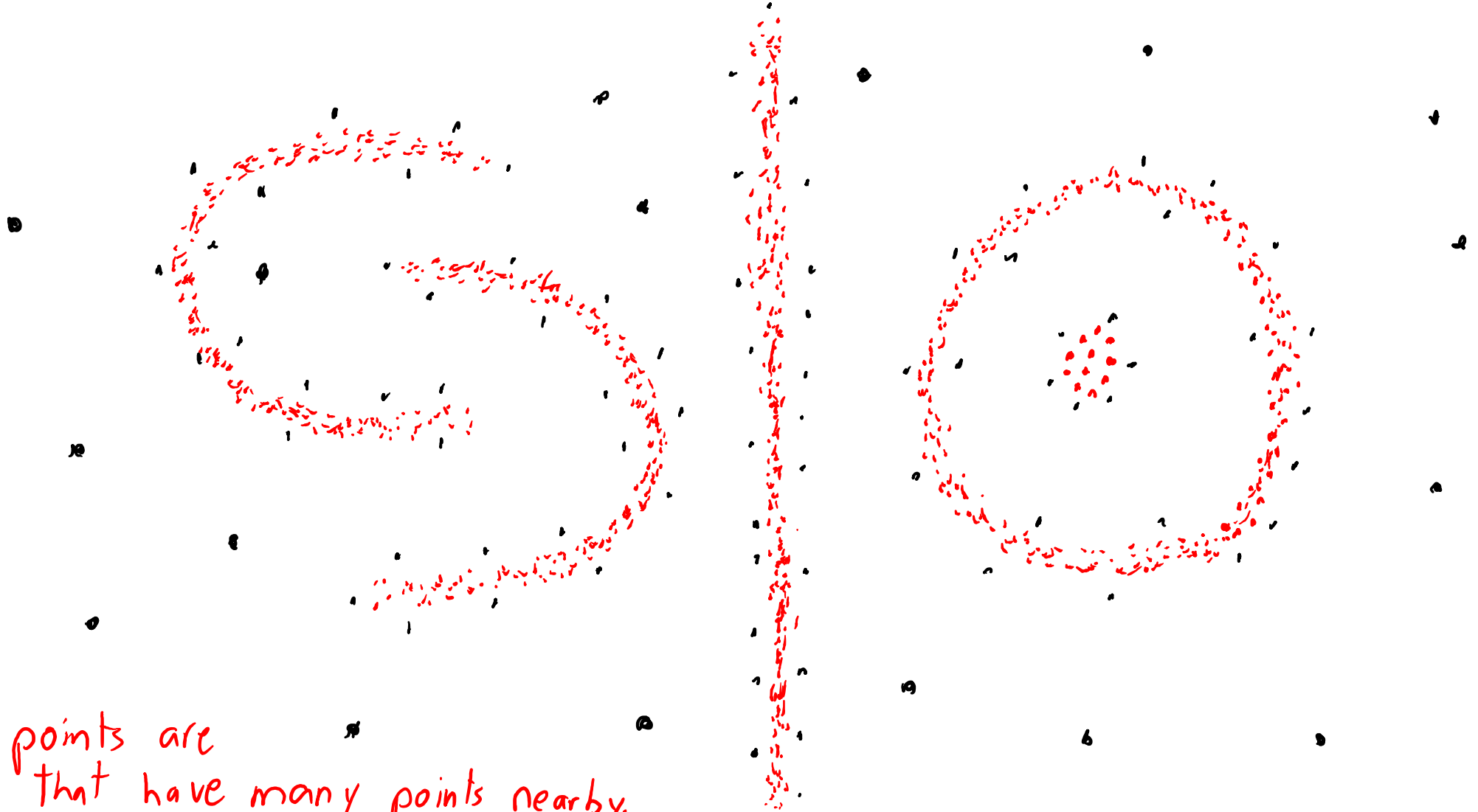


E.g., if $\text{minNeighbours} = 3$
then this is a “core”
point since 6 points are
“neighbours”

Density-Based Clustering

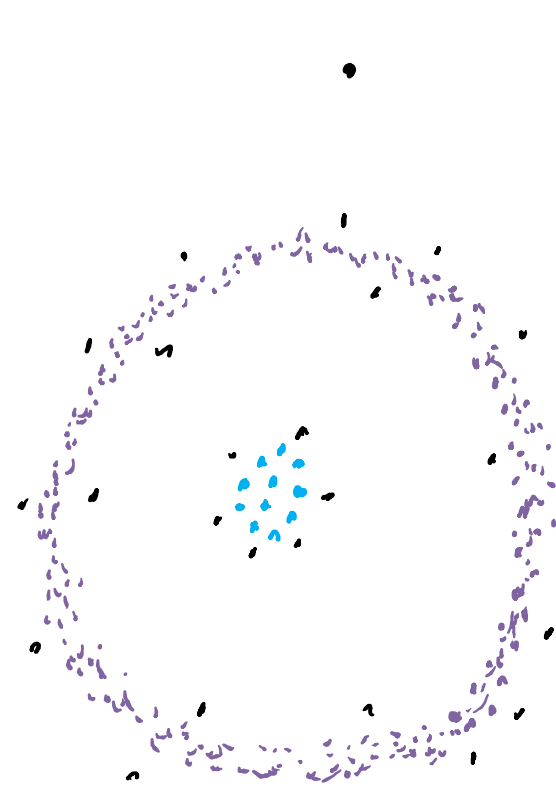
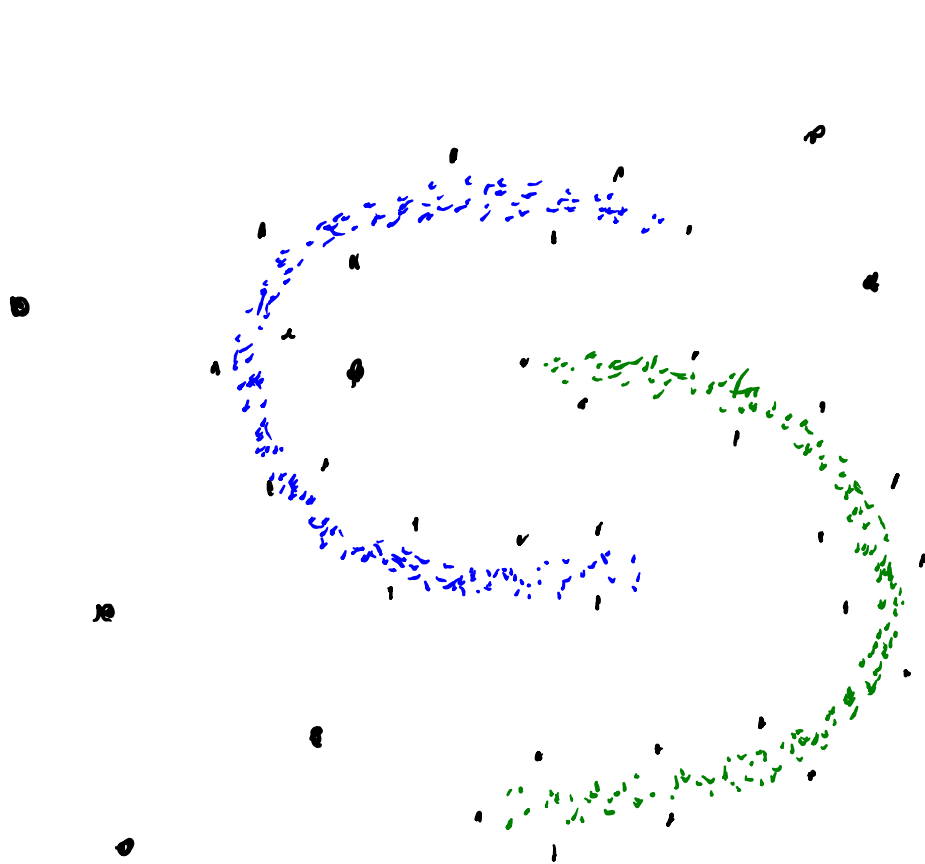


Density-Based Clustering



"Core" points are points that have many points nearby.

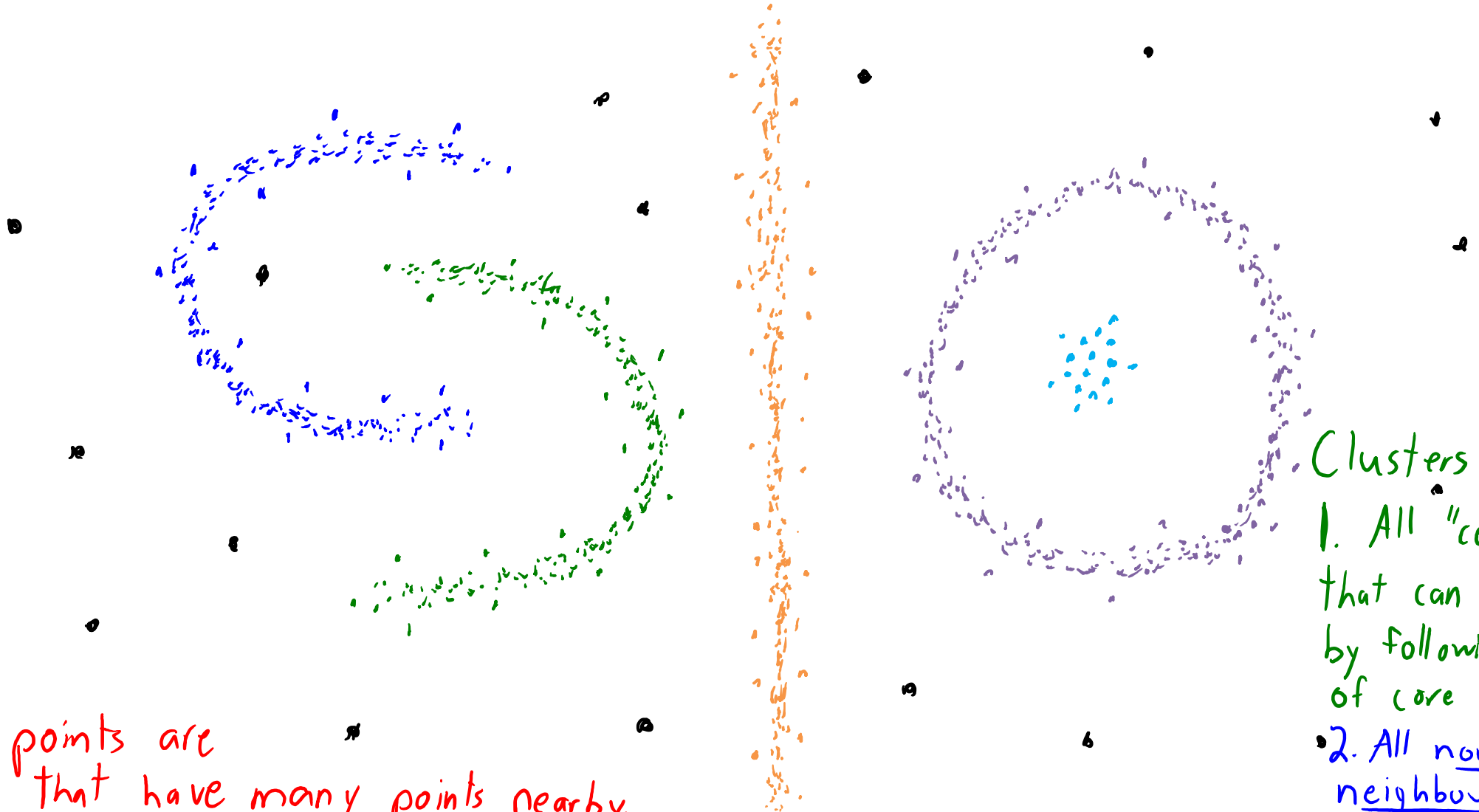
Density-Based Clustering



Clusters contain:
1. All "core" points that can be reached by following a sequence of core points.

"Core" points are points that have many points nearby.

Density-Based Clustering



"Core" points are points that have many points nearby.

Clusters contain:

1. All "core" points that can be reached by following a sequence of core points.
2. All non-core neighbours of core points (boundary points)

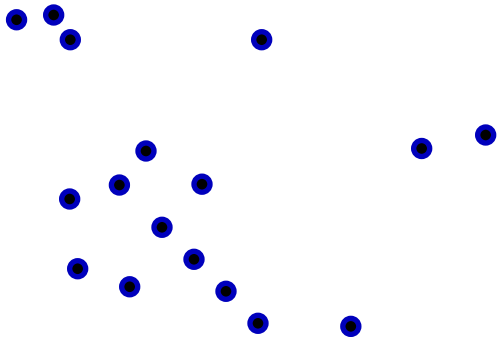
Density-Based Clustering Pseudo-Code

- Intuitively, density-based clustering **algorithm** implements a “chain reaction” throughout the dense areas.
- For each example x_i :
 - If x_i is already assigned to a cluster, do nothing.
 - Else: Test whether x_i is a ‘core’ point ($\geq \text{minNeighbours}$ examples within ‘ ϵ ’, including x_i)
 - If x_i is not core point, do nothing (this could be an outlier).
 - If x_i is a core point, make a **new cluster** and call the “**expand cluster**” function.
 - Which spreads the “reaction” to nearby points.

Density-Based Clustering Pseudo-Code

- “Expand cluster” function:
 - Assign to this cluster all x_j within distance ‘ ϵ ’ of core point x_i to this cluster.
 - For each new “core” point found, call “expand cluster” (recursively).

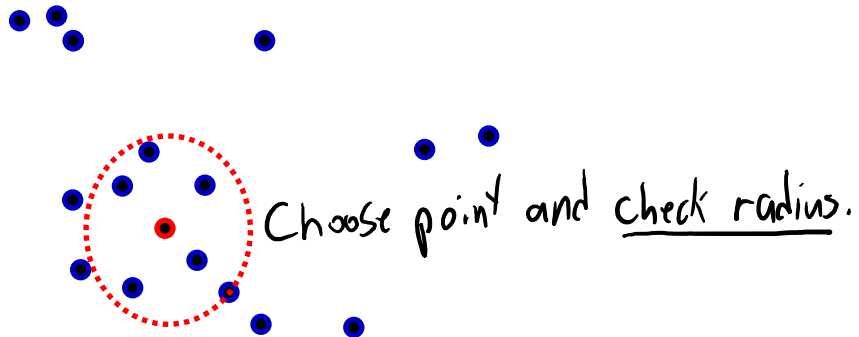
minNeighbors = 3



Density-Based Clustering Pseudo-Code

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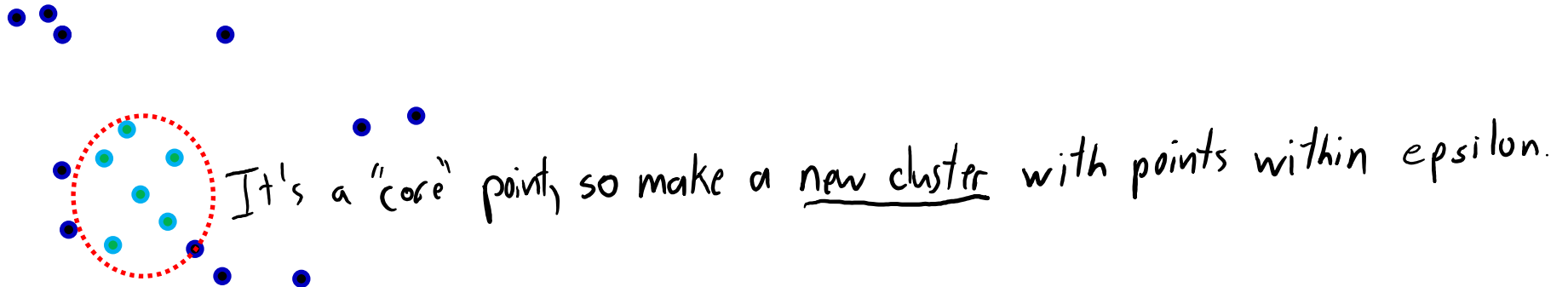
minNeighbors = 3



Density-Based Clustering Pseudo-Code

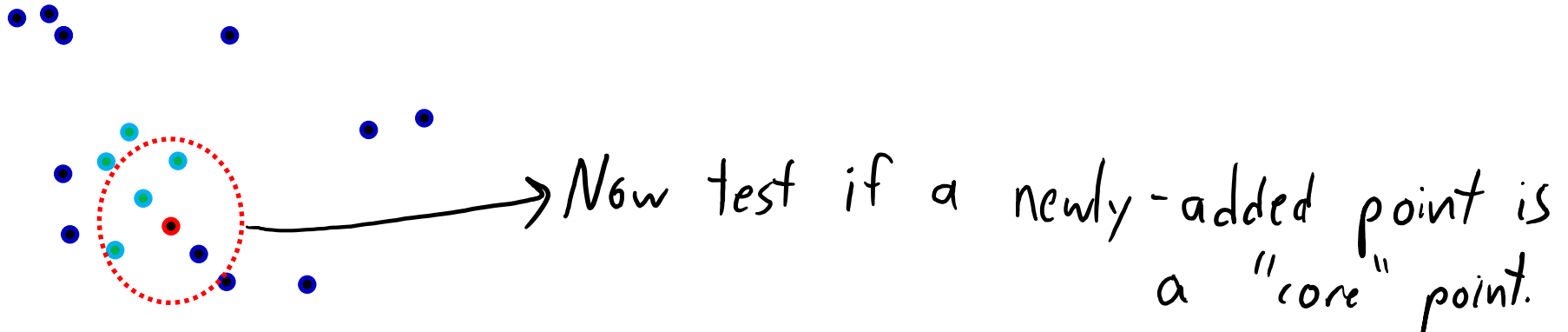
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minNeighbors = 3



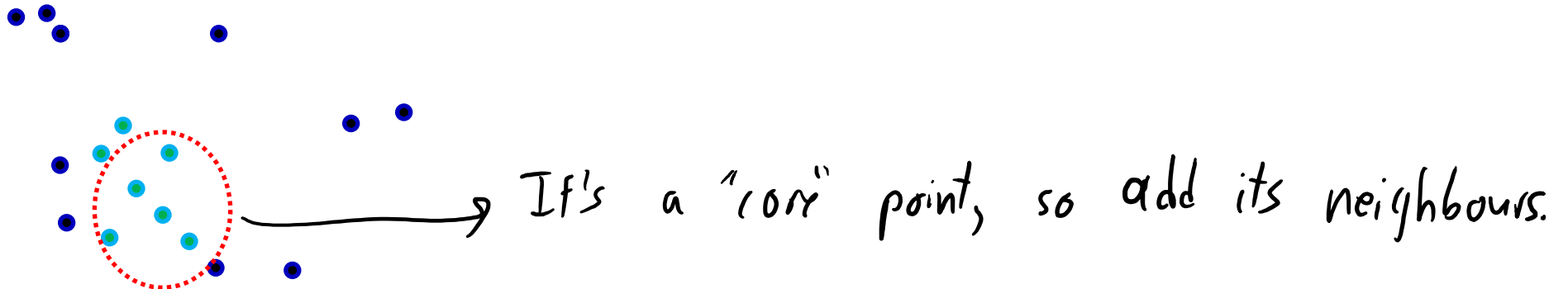
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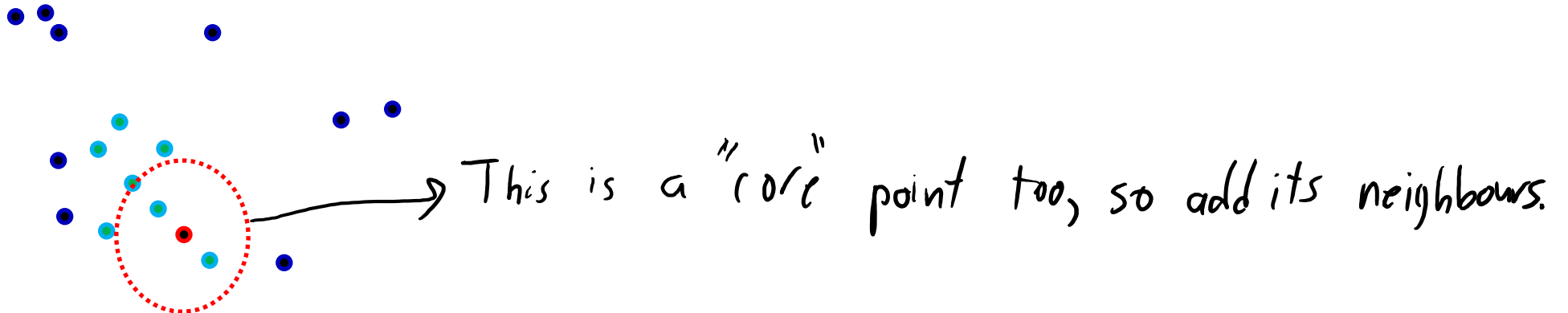
Density-Based Clustering Pseudo-Code

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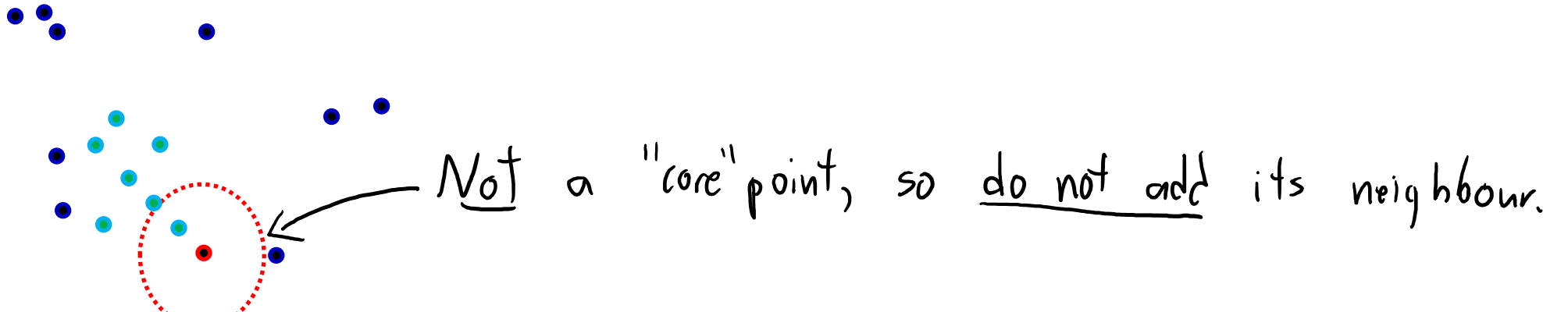
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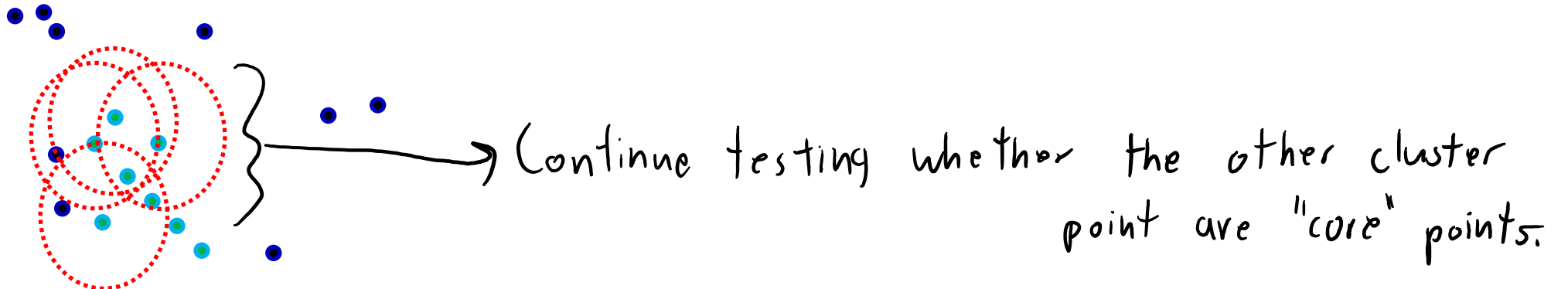
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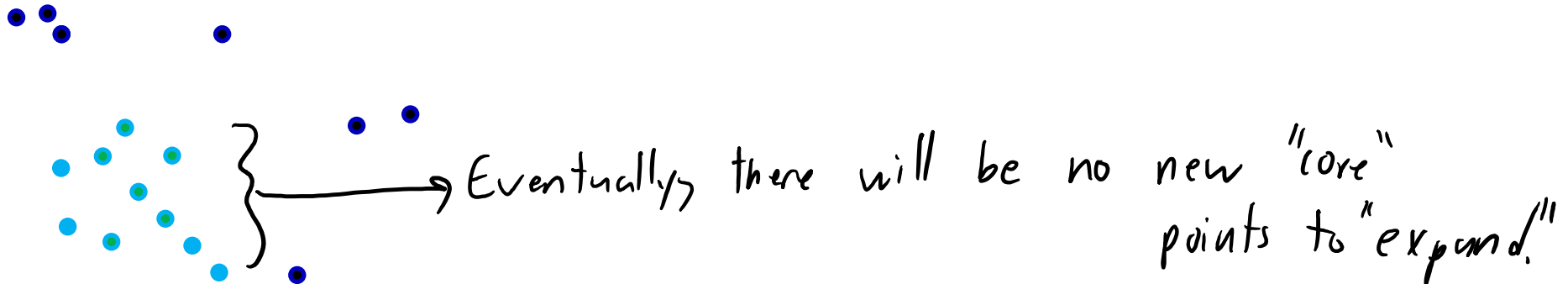
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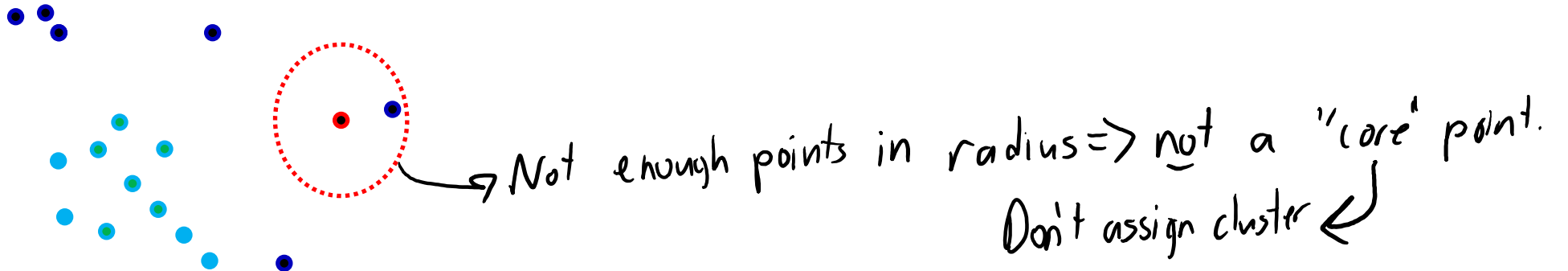
Density-Based Clustering Pseudo-Code

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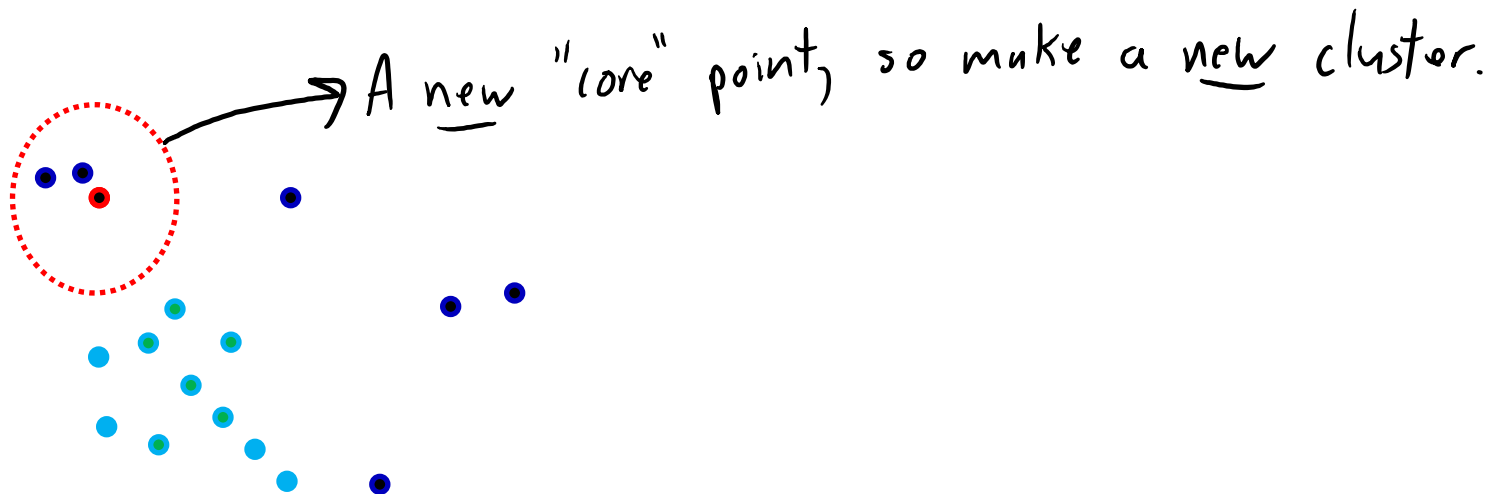
Density-Based Clustering Pseudo-Code

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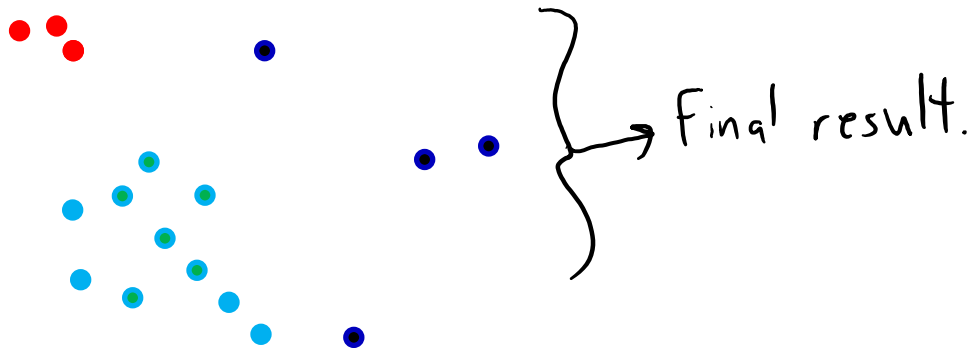
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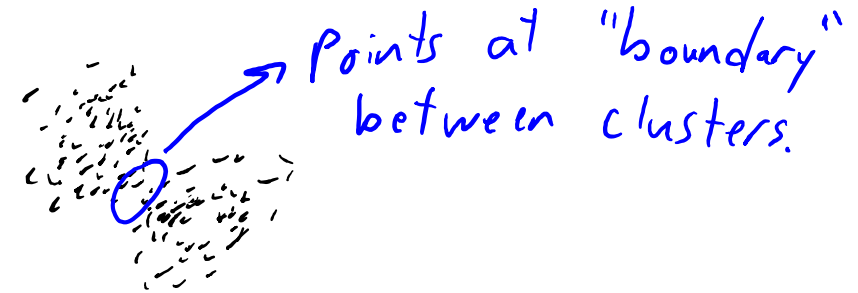
Density-Based Clustering Pseudo-Code

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 - For each new “core” point found, call “expand cluster” (recursively).



Density-Based Clustering Issues

- Some points are not assigned to a cluster.
 - Good or bad, depending on the application
- Ambiguity of “non-core” (boundary) points:
 - Otherwise, not sensitive to initialization (except for boundary points).
- Sensitive to the choice of ϵ and minNeighbours.
 - Original paper proposed an “elbow” method (see bonus slide).
- If you get a new example, finding cluster is expensive.
 - Need to compute distances to core points (or maybe all training points).
- In high-dimensions, need a lot of points to ‘fill’ the space.



DBSCAN Cost

- Cost of DBSCAN
 - Worst Case: $O(dN^2)$
 - For each point, needs to calculate distance to every other point to see if within ϵ
 - Often can be sped up to $O(n \log n)$ 

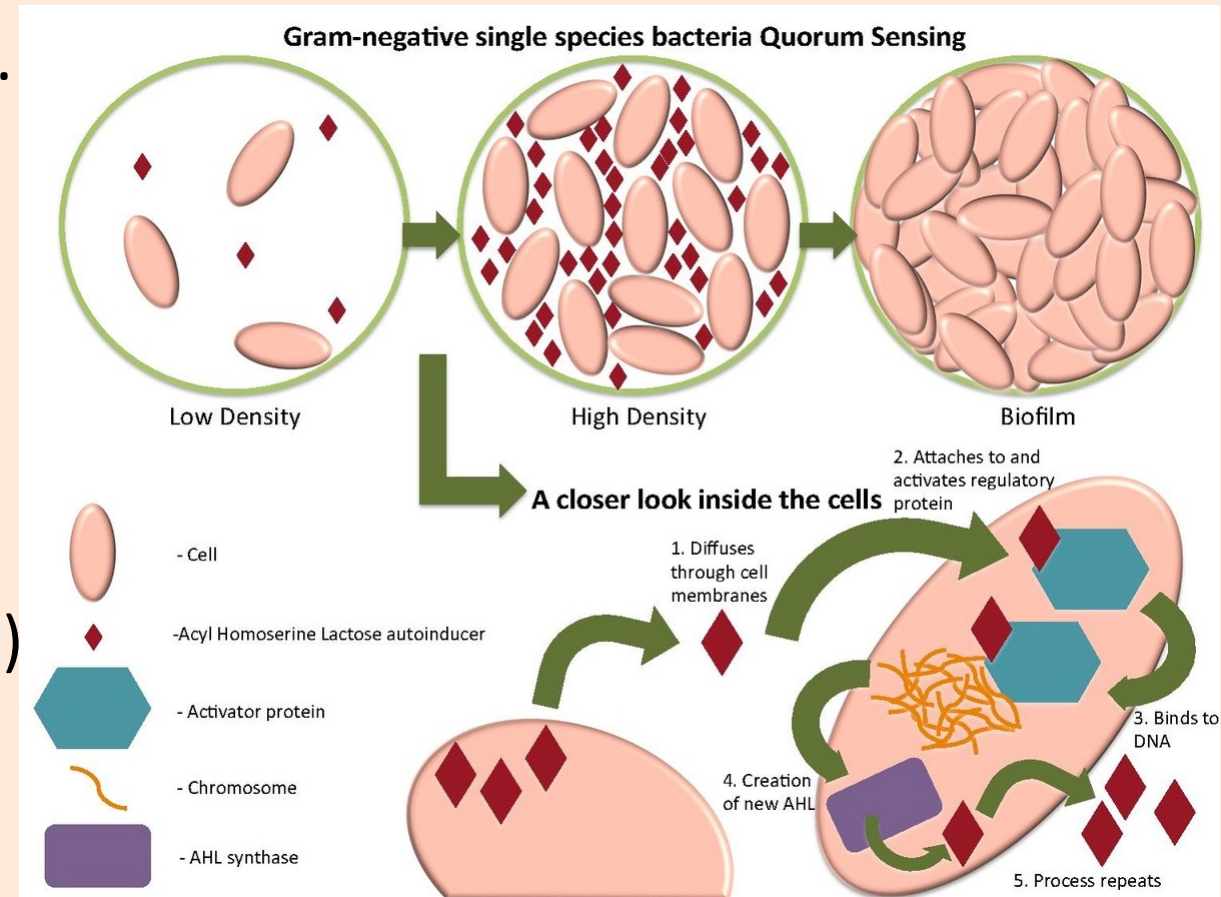
Complexity [\[edit \]](#)

DBSCAN visits each point of the database, possibly multiple times (e.g., as candidates to different clusters). For practical considerations, however, the time complexity is mostly governed by the number of regionQuery invocations. DBSCAN executes exactly one such query for each point, and if an [indexing structure](#) is used that executes a [neighborhood query](#) in $O(\log n)$, an overall average runtime complexity of $O(n \log n)$ is obtained (if parameter ϵ is chosen in a meaningful way, i.e. such that on average only $O(\log n)$ points are returned). Without the use of an accelerating index structure, or on degenerated data (e.g. all points within a distance less than ϵ), the worst case run time complexity remains $O(n^2)$. The $\binom{n}{2} - n = (n^2 - n)/2$ -sized upper triangle of the distance matrix can be materialized to avoid distance recomputations, but this needs $O(n^2)$ memory, whereas a non-matrix based implementation of DBSCAN only needs $O(n)$ memory.

<https://en.wikipedia.org/wiki/DBSCAN>

Density-Based Clustering in Bacteria

- Quorum sensing:
 - Bacteria continuously release a particular molecule.
 - They have sensors for this molecule.
- If sensors become very active:
 - It means cell density is high.
 - Causes cascade of changes in cells.
(Some cells “stick together” to form a physical cluster via “biofilm”).



Density-Based Clustering in People

- “High density leading a chain reaction” can also happen in people.

COVID-19 cluster reported at Port Coquitlam Costco; outbreak at VGH declared

BY DEAN RECKSIDLER AND HANA MAE NASSAR

Posted Mar 15, 2021 9:38 am PDT Last Updated Mar 16, 2021 at 1:09 am PDT



- “Social distancing”: try to reduce the number of people within ϵ .
- “Wearing masks”: try to increase the ϵ needed for a chain reaction.

DB Scan is big and popular way for clustering :)

Next Topic: Ensemble Clustering

Ensemble Clustering

? question ☆

stop following 23 views

Multiple random runs of K means

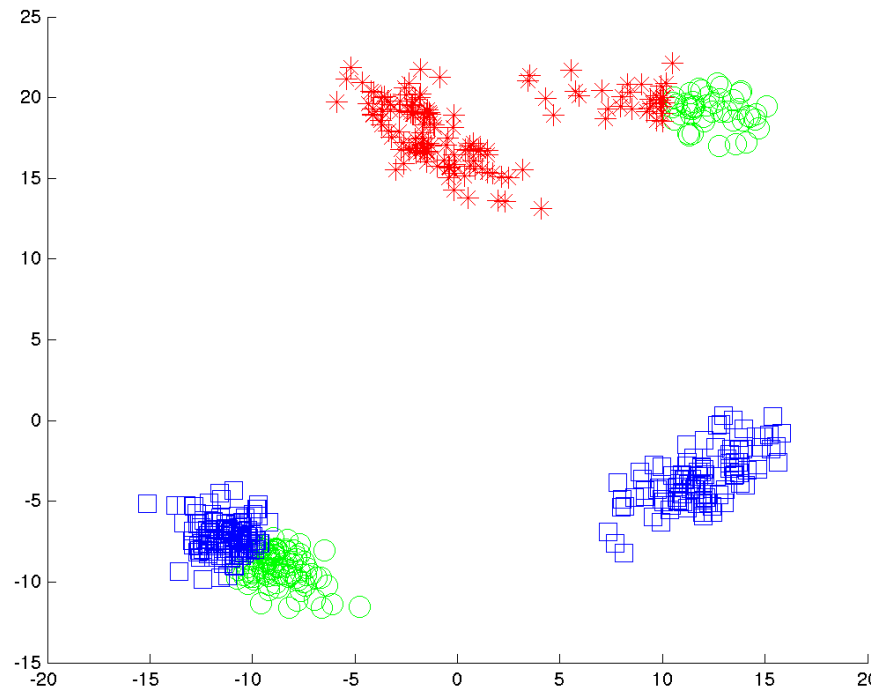
I was wondering how running K Means (original version, not K means ++) several times with random initializations can help us make an accurate model. K Means outputs the class labels of all the samples. We definitely can't use mode of all the labels it got in different runs because class labels from different runs don't make any sense when compared. We somehow have to see what points are coming in the same cluster in a lot of runs..I am not sure, how do we do it?

- We can consider **ensemble methods** for clustering.
 - “Consensus clustering”
- It's a good/important idea:
 - **Bootstrapping** is widely-used.
 - “Do clusters change if the data was slightly different?”
- But we **need to be careful** about how we combine models.

So far, talked about picking different random points for clustering and minimize f
This is another way

Ensemble Clustering

- E.g., run k-means 20 times and then cluster using the mode of each $\hat{y}_i$.
- Normally, averaging across models doing different things is good.

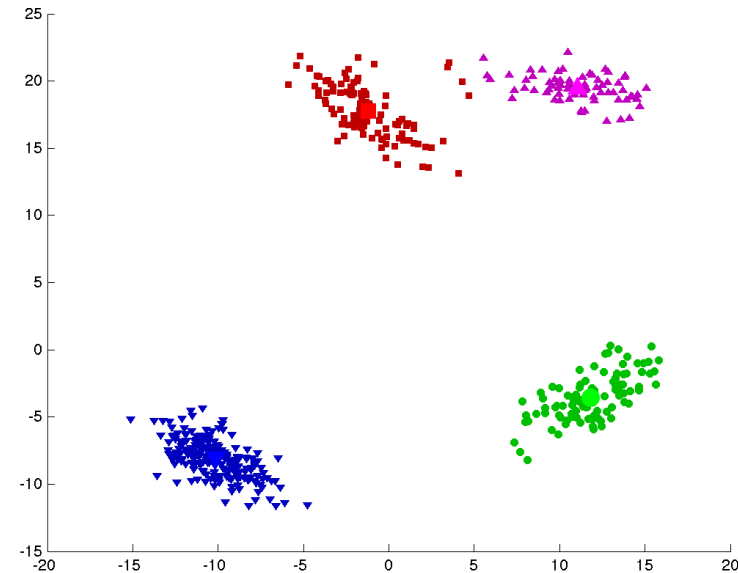
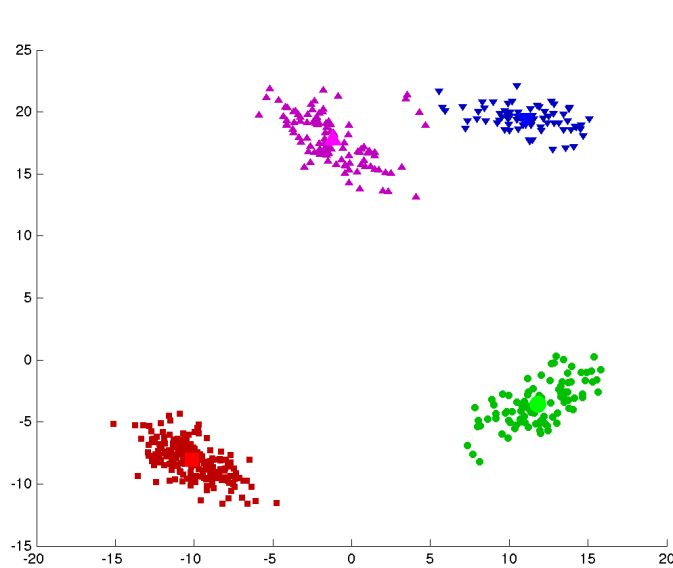


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- But this is a bad ensemble method: **worse than k-means on its own.**

Label Switching Problem

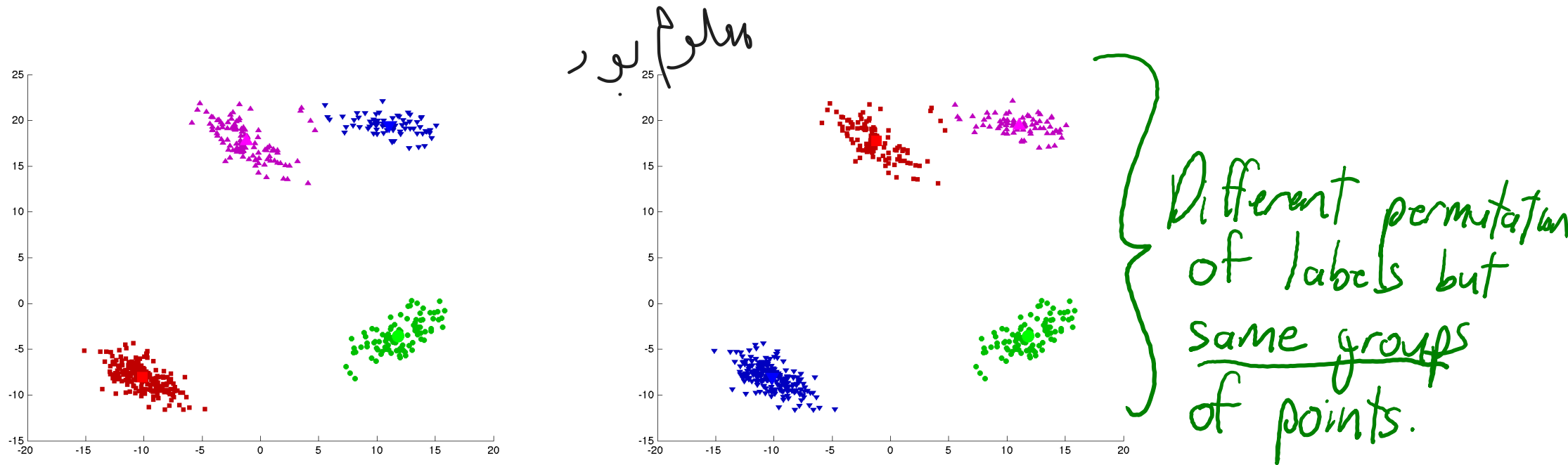
- This doesn't work because of “label switching” problem:
 - The cluster labels $\hat{y}_i$ are meaningless.
 - We could get same clustering with permuted labels (“exchangeable”):



- All $\hat{y}_i$ become equally likely as number of initializations increases.

Addressing Label Switching Problem

- Ensembles can't depend on label "meaning":
 - Don't ask "is point x_i in red square cluster?", which is meaningless.
 - Ask "is point x_i in the same cluster as x_j ?", which is meaningful.



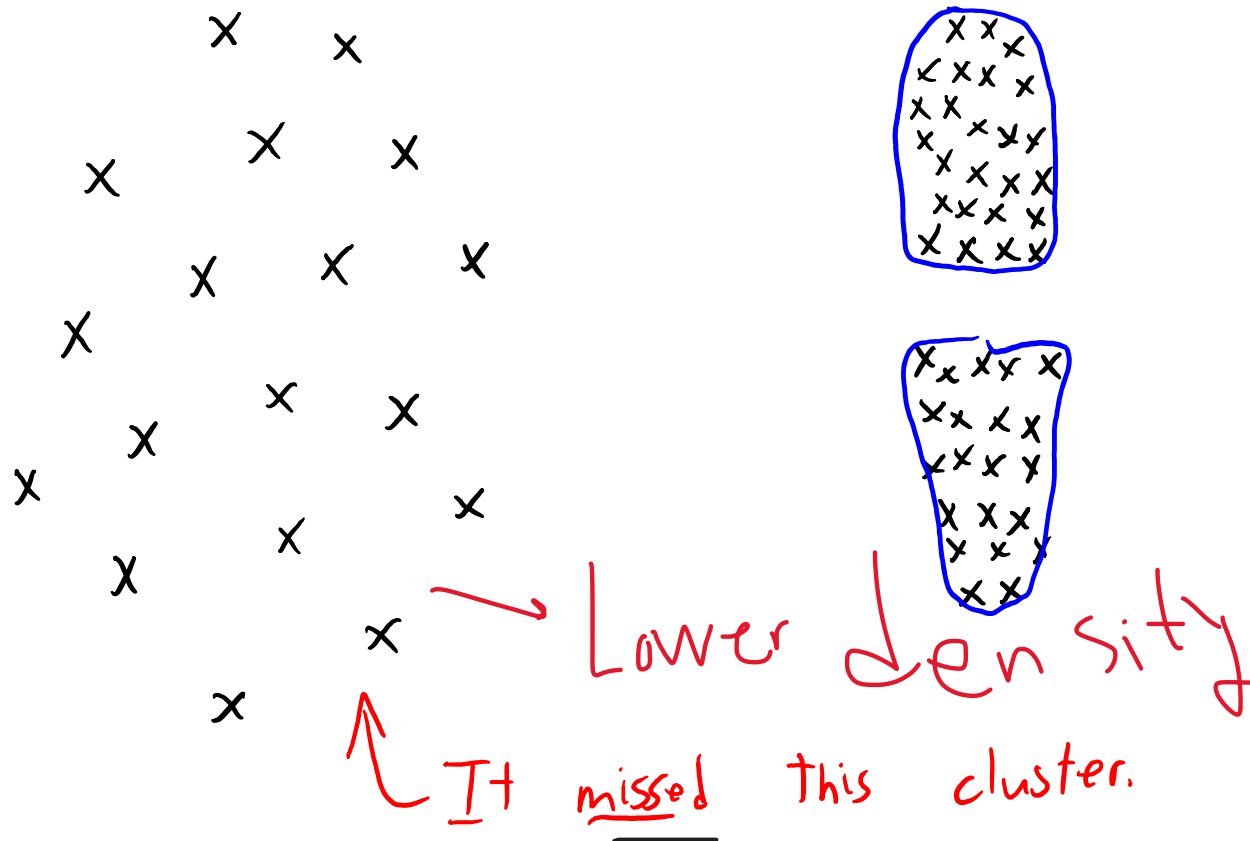
- Bonus slides give an example method ("UBClustering").

Next Topic: Hierarchical Clustering

DB Scan

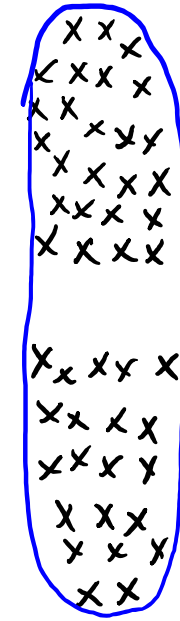
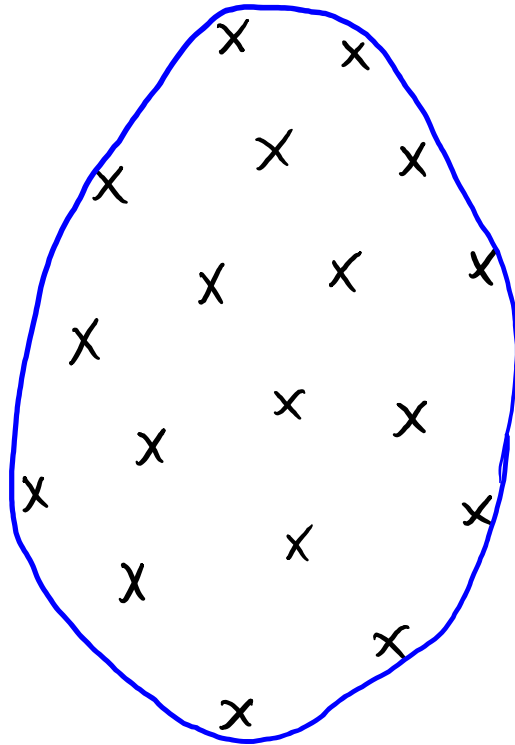
Differing Densities

- Consider density-based clustering on this data:



Differing Densities

- Increase **epsilon** and run it again:

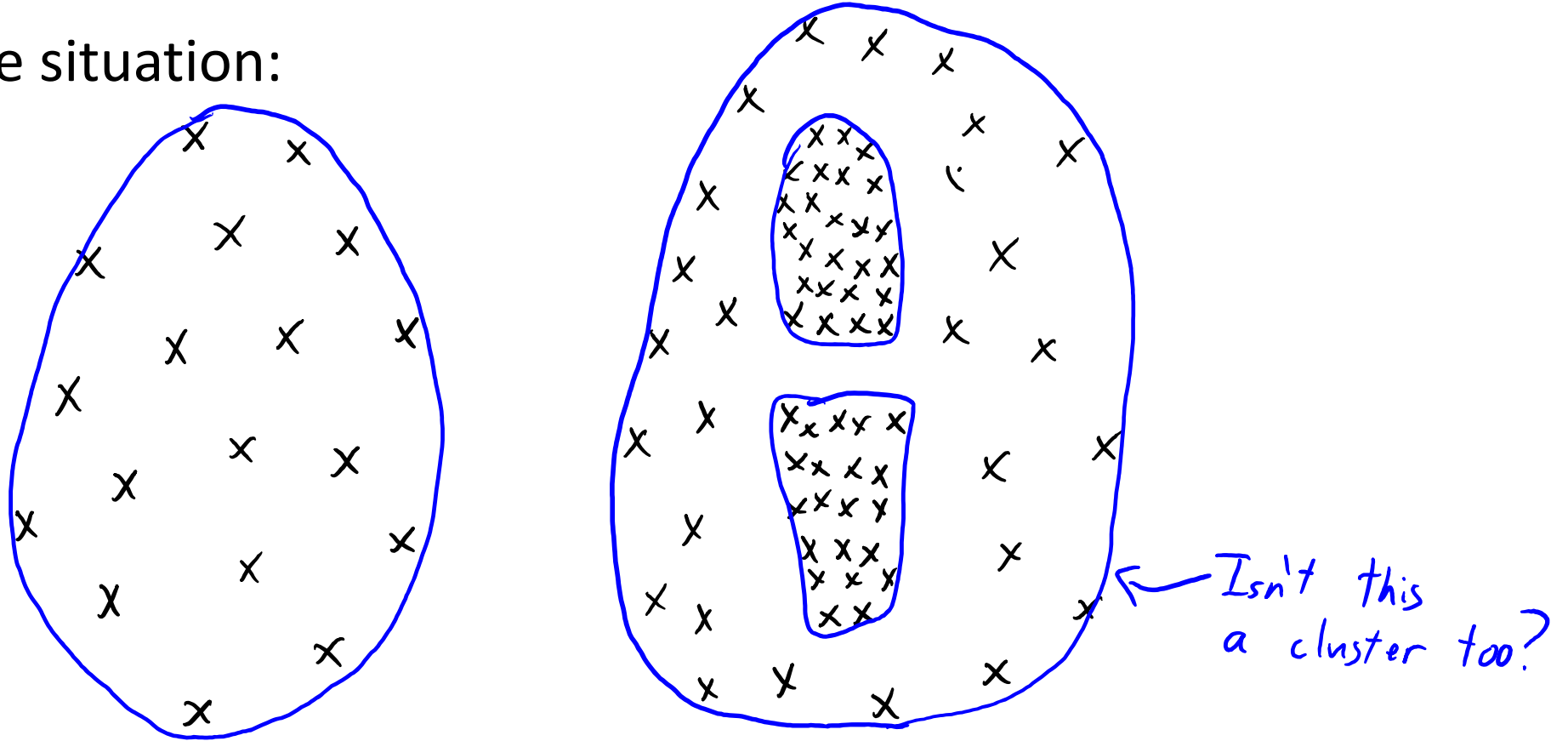


These 2 clusters
are now "close."

- There may be **no density-level** that gives you 3 clusters.

Differing Densities

- Here is a worse situation:

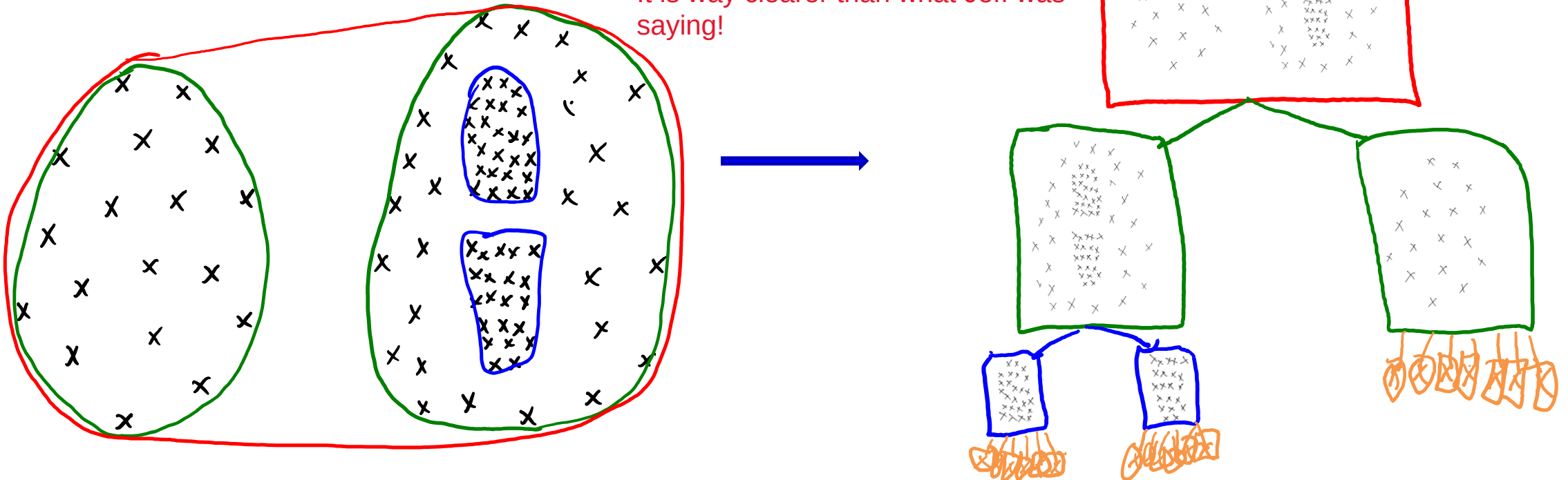


- Now you need to choose between coarse/fine clusters.
- Instead of fixed clustering, we often want **hierarchical clustering**.

Hierarchical Clustering

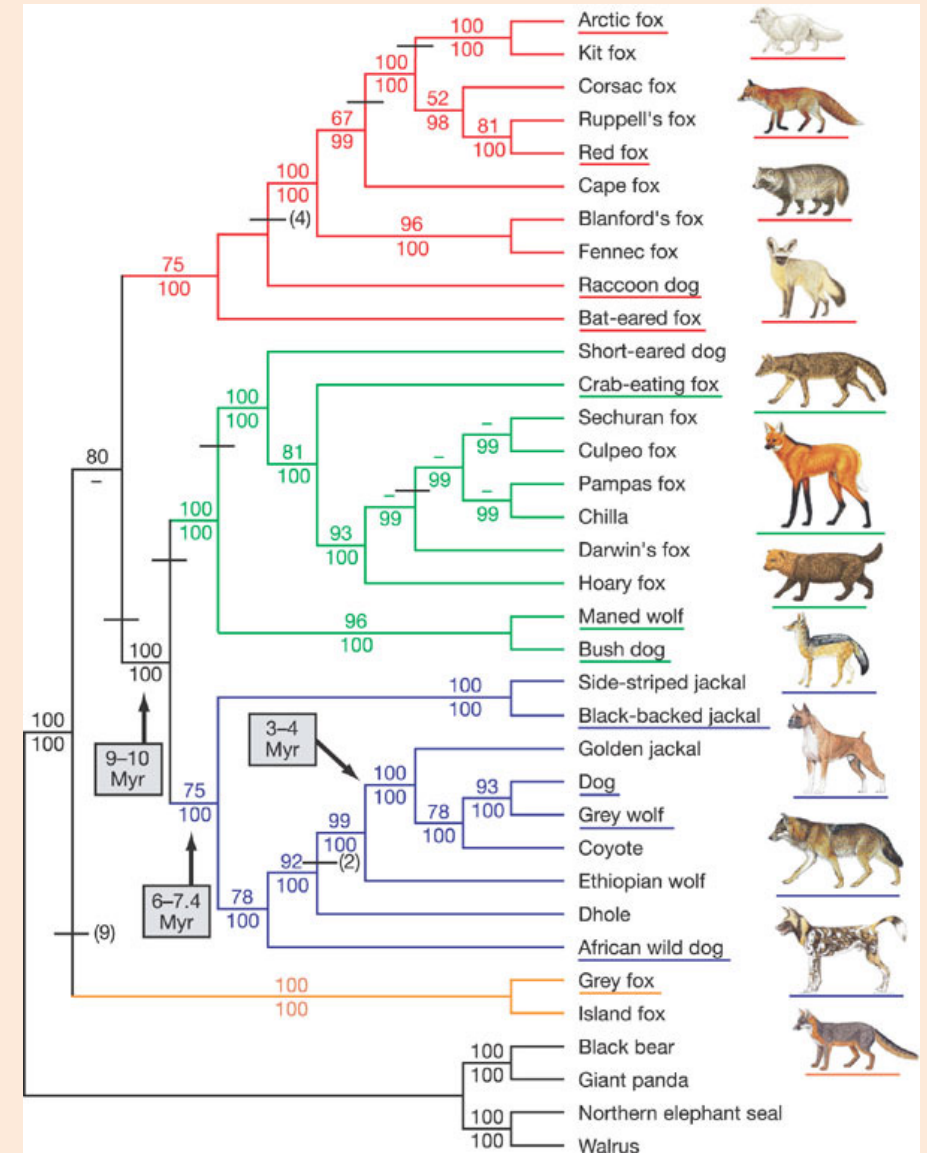
- Hierarchical clustering produces a tree of clusterings.
 - Each node in the tree splits the data into 2 or more clusters.
 - Much more information than using a fixed clustering.
 - Often have individual data points as leaves.

It is way clearer than what Jeff was saying!



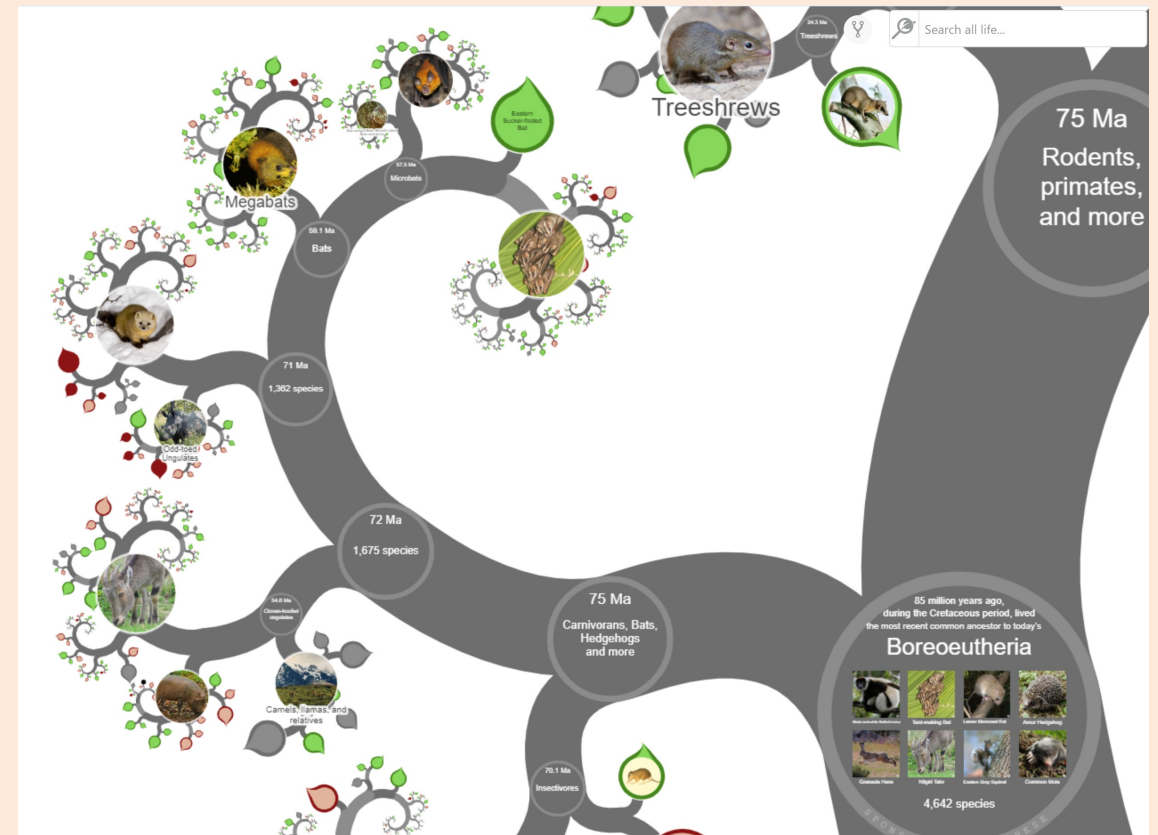
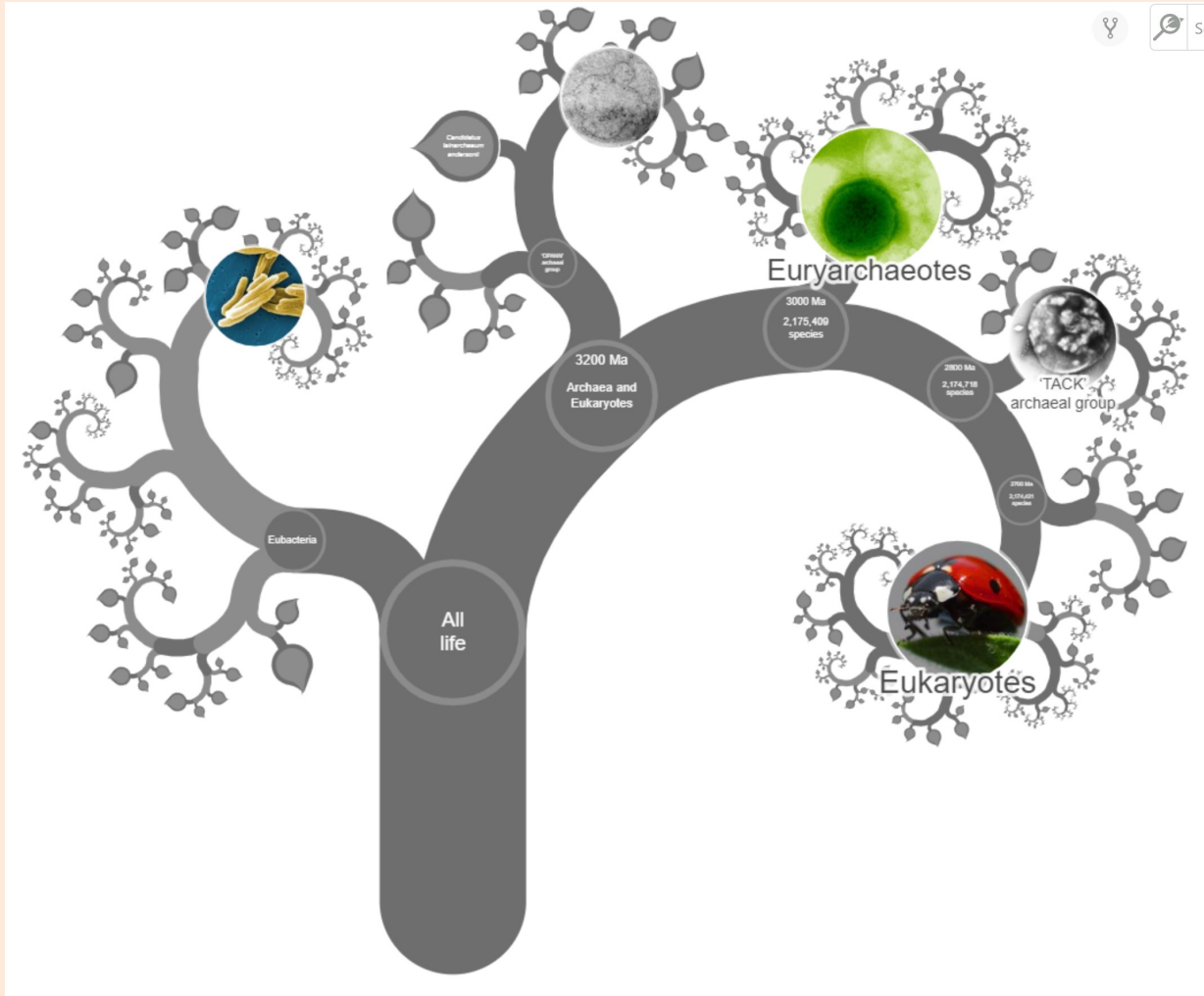
Application: Phylogenetics

- We sequence genomes of a set of organisms.
- Can we construct the “tree of life”?
- Comments on this application:
 - On the right are individuals.
 - As you go left, clusters merge.
 - Merges are ‘common ancestors’.
- More useful information in the plot:
 - Line lengths: chosen here to approximate time.
 - Numbers: #clusterings across bootstrap samples.
 - ‘Outgroups’ (walrus, panda) are a sanity check.



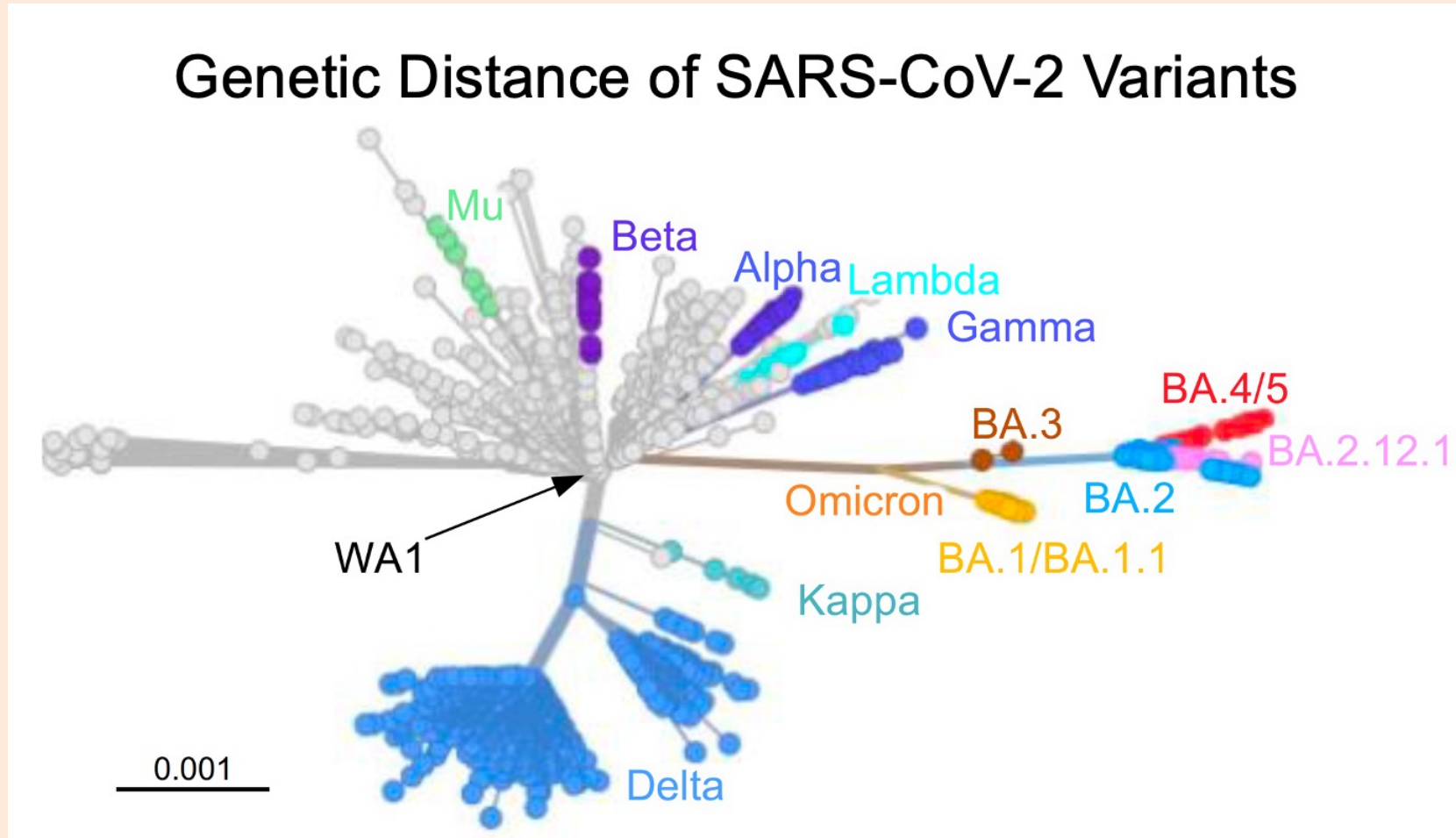
Application: Phylogenetics

- Interactive demo of model of full tree of life: www.onezoom.org



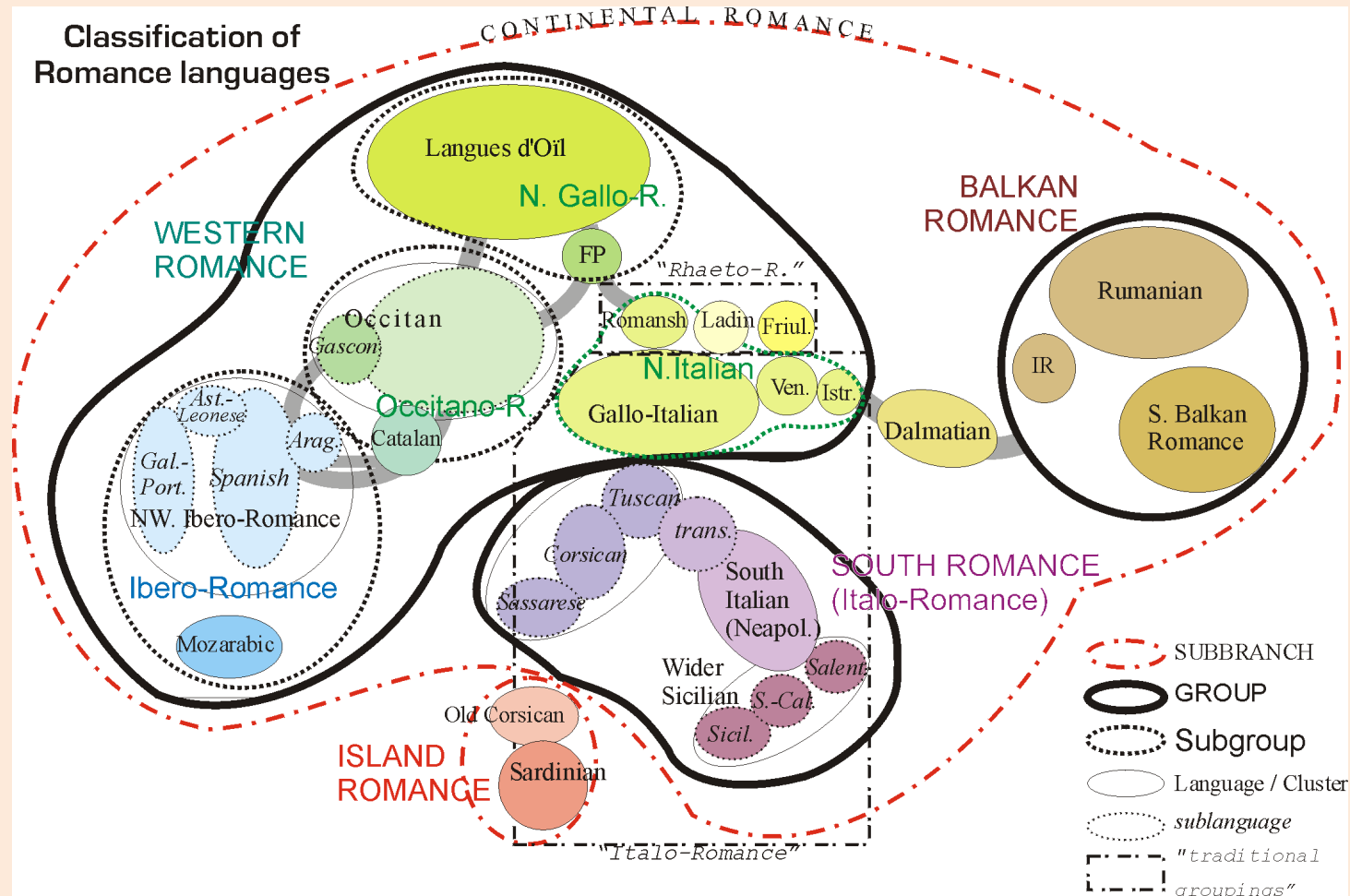
Application: Phylogenetics

- Model of Covid-19 variants:



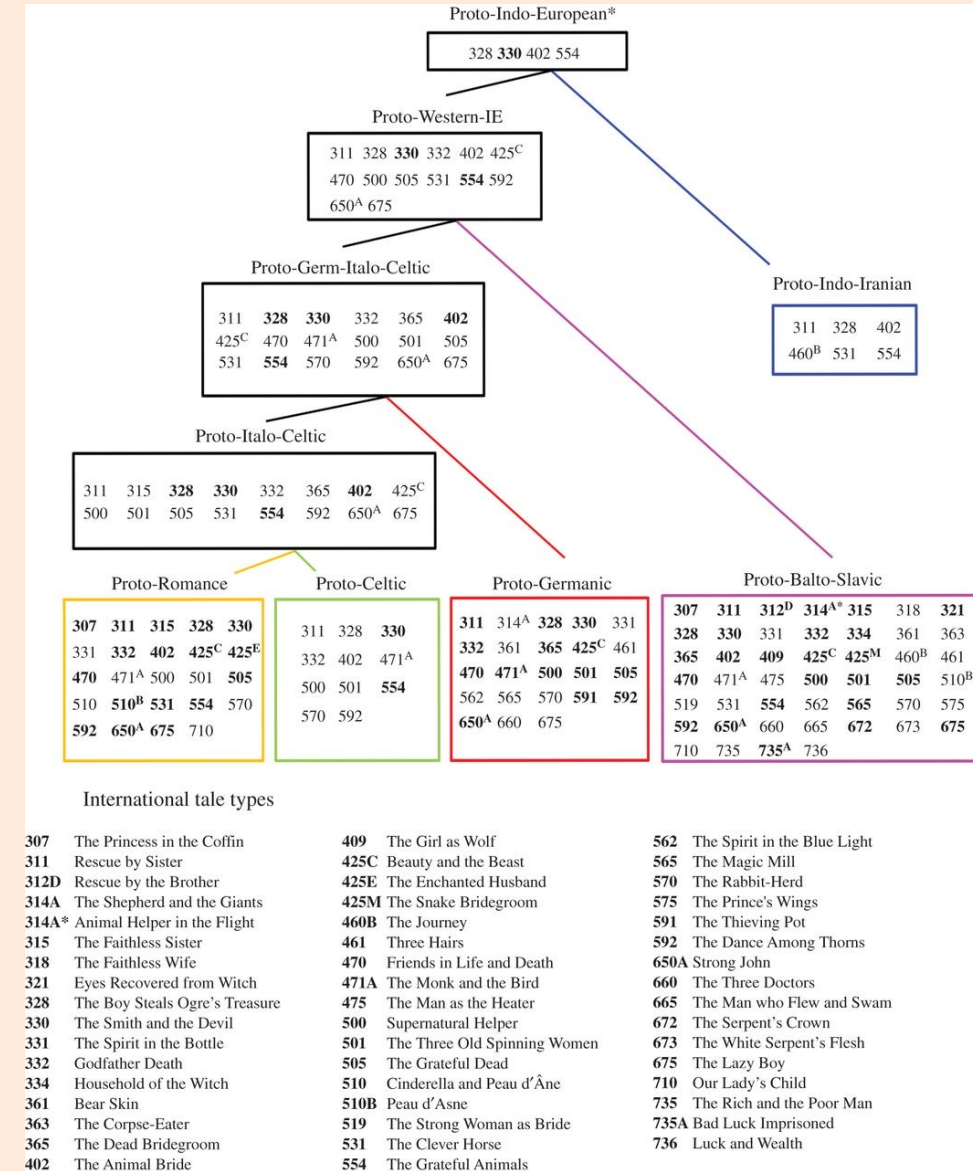
Application: Phylogenetics

- Comparative method in linguistics studies evolution of languages:



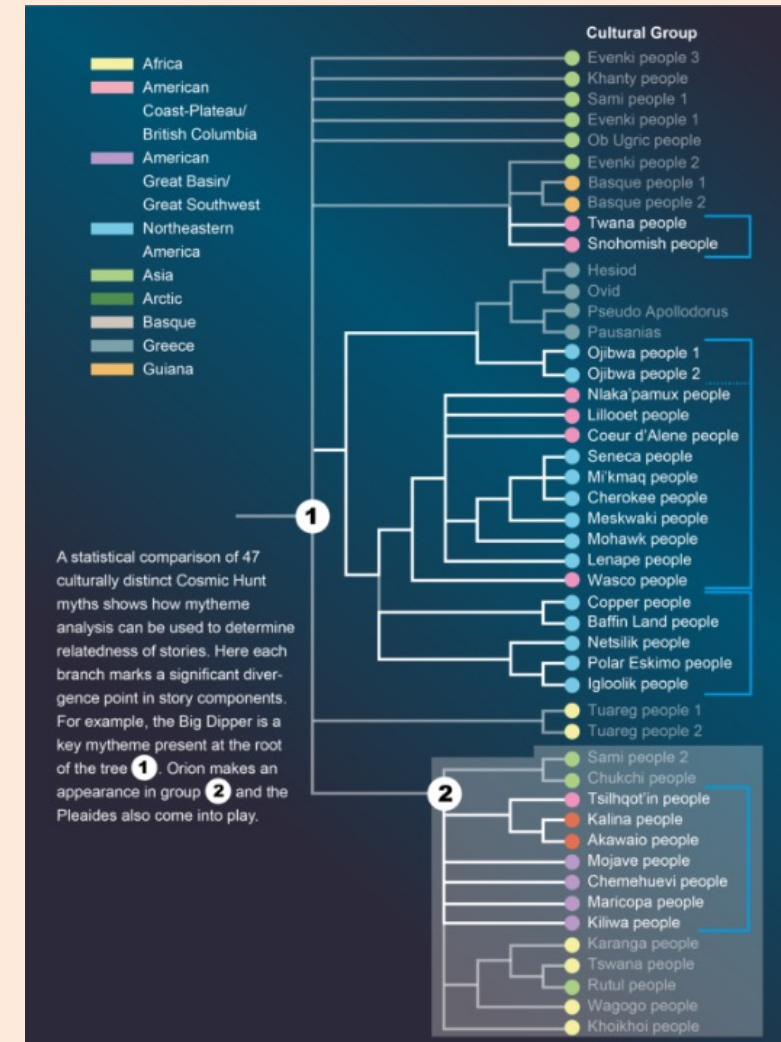
Application: Phylogenetics

- January 2016: evolution of fairy tales.
 - Evidence that “Devil and the Smith” goes back to bronze age.
 - “Beauty and the Beast” published in 1740, but might be 2500-6000 years old.



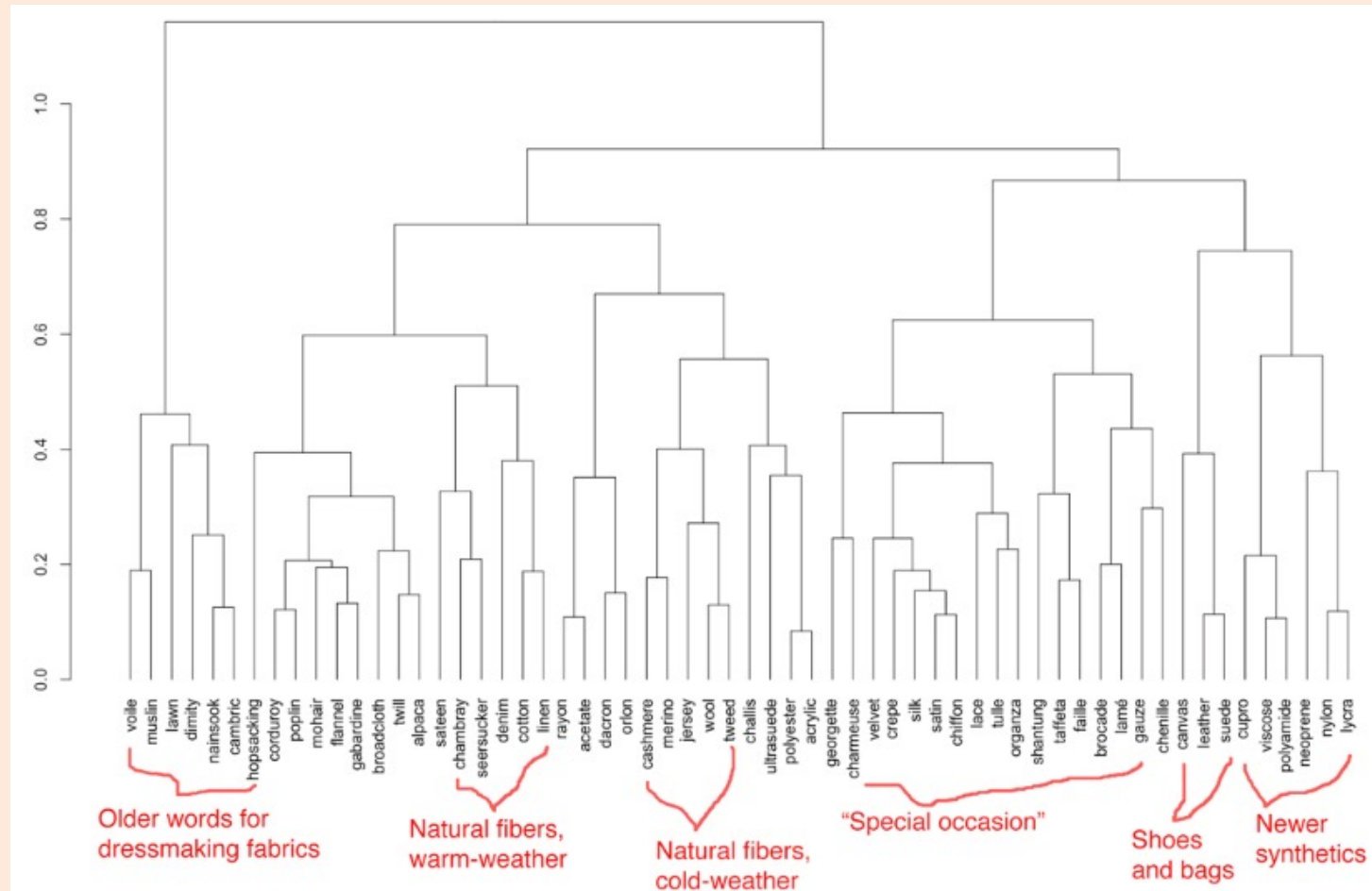
Application: Phylogenetics

- January 2016: evolution of fairy tales.
 - Evidence that “Devil and the Smith” goes back to bronze age.
 - “Beauty and the Beast” published in 1740, but might be 2500-6000 years old.
- September 2016: evolution of myths.
 - “Cosmic hunt” story:
 - Person hunts animal that becomes constellation.
 - Previously known to be at least 15,000 years old.
 - May go back to paleolithic period.



Application: Fashion?

- Hierarchical clustering of clothing material words in Vogue:

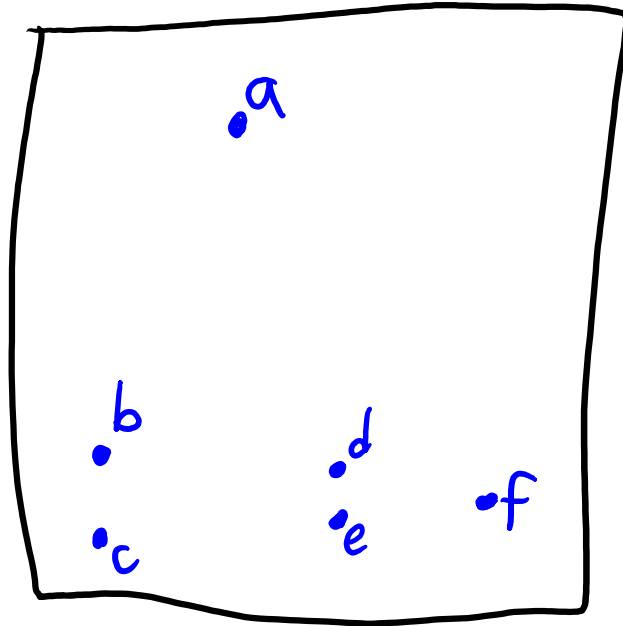


Agglomerative (Bottom-Up) Clustering



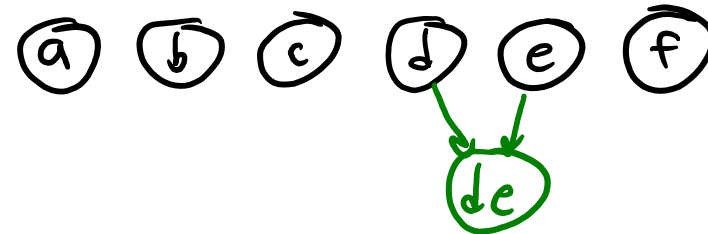
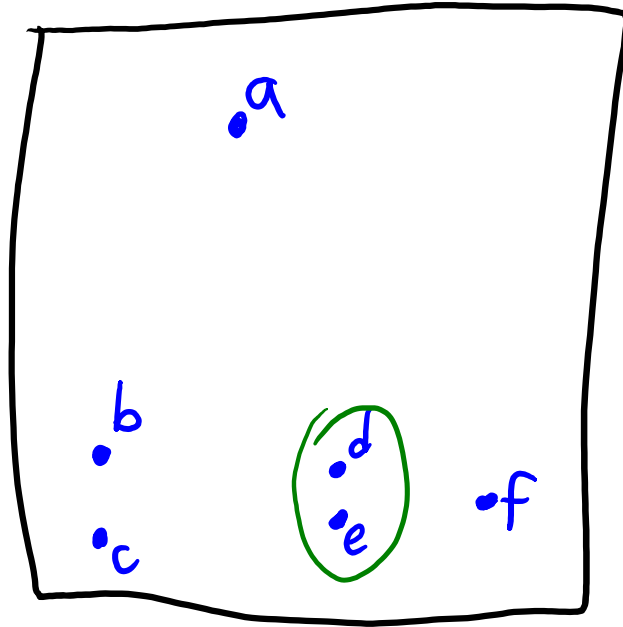
One type of Hierarchical Clustering

- Most common hierarchical method: **agglomerative clustering**.
 1. Starts with **each point in its own cluster**.



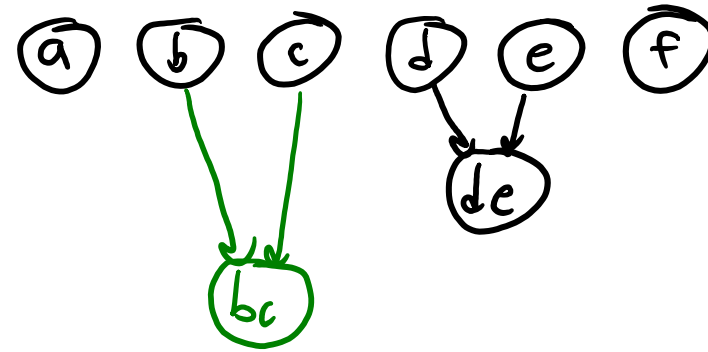
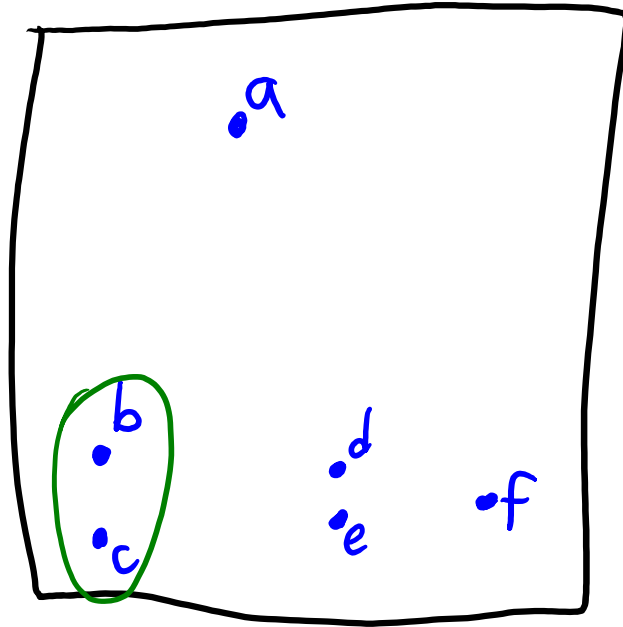
Agglomerative (Bottom-Up) Clustering

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 2. Each step **merges the two “closest” clusters**.



Agglomerative (Bottom-Up) Clustering

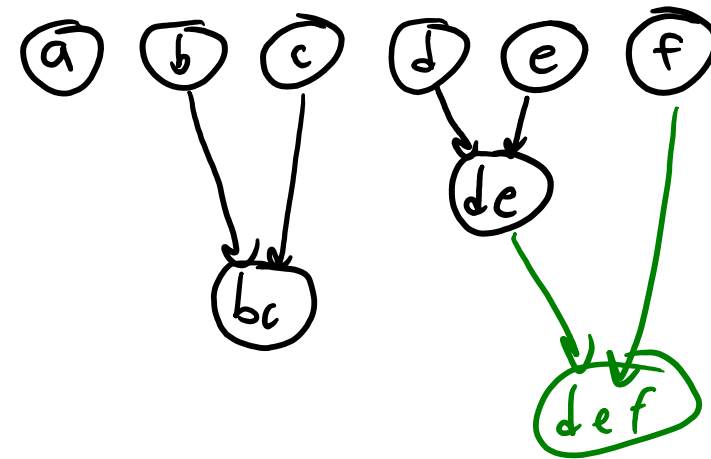
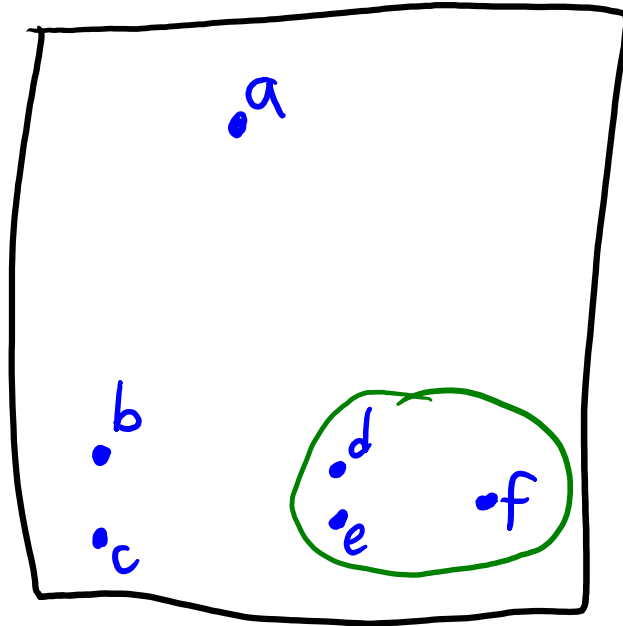
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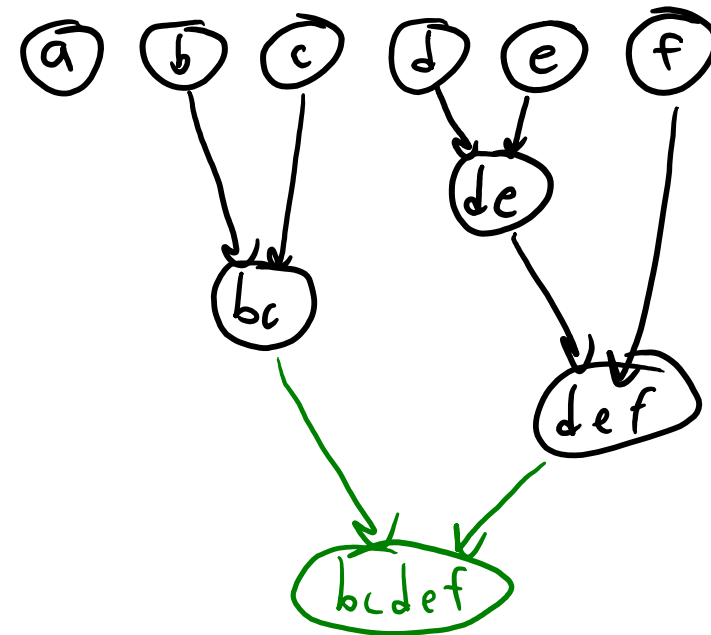
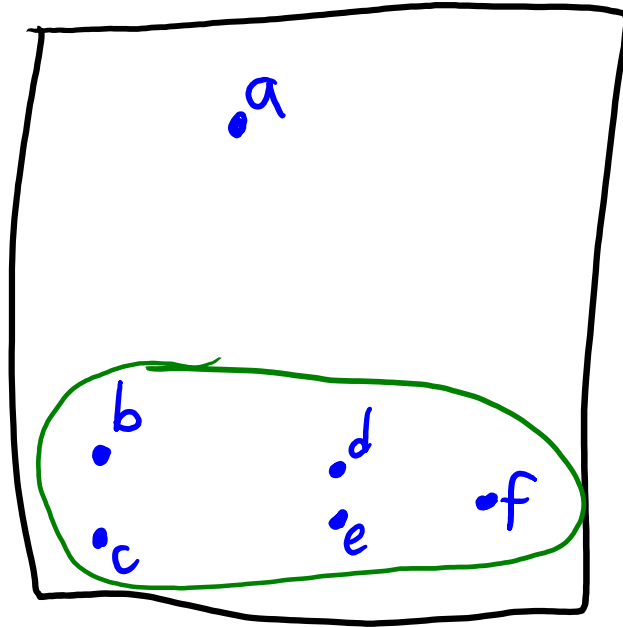
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All of them have some distance calculated and then set some threshold



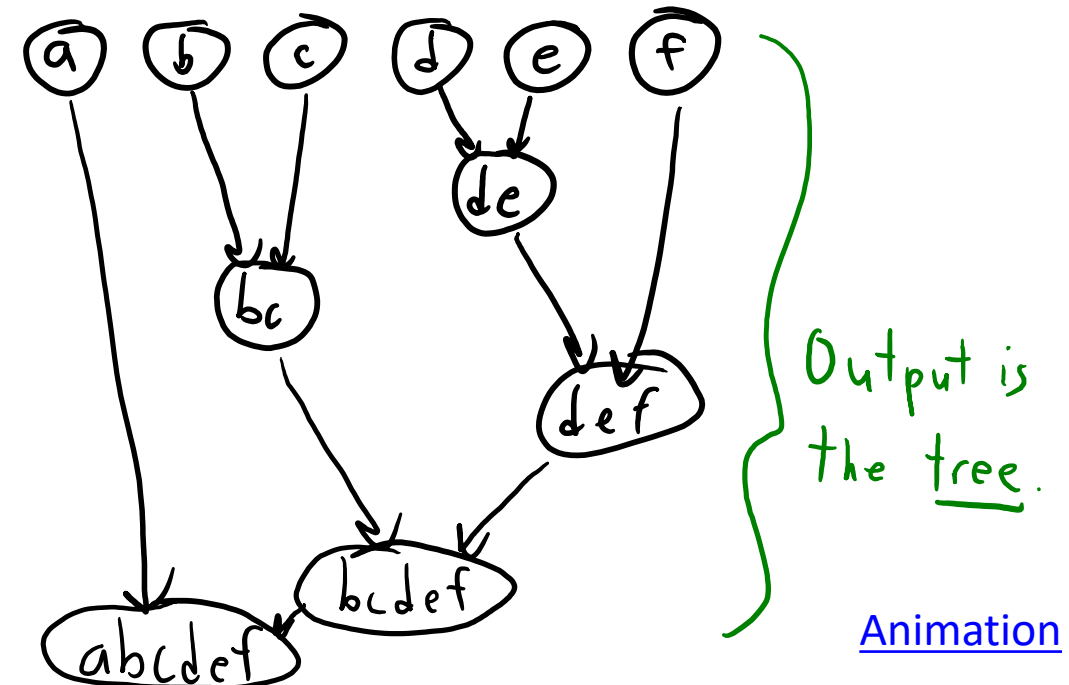
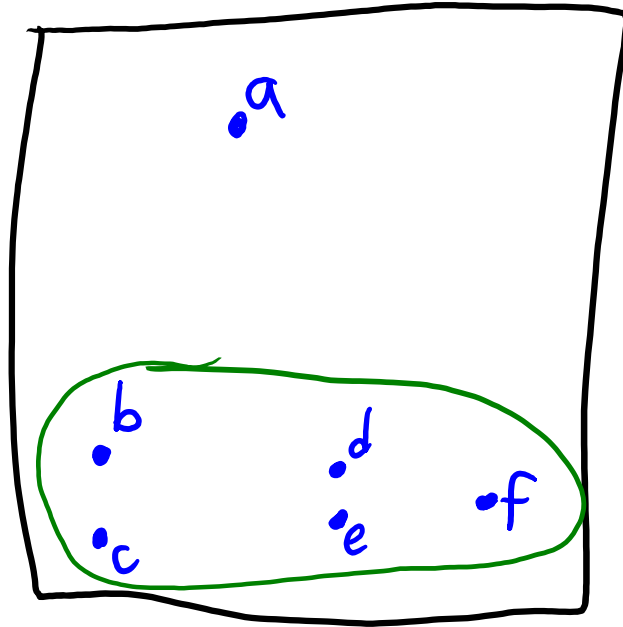
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Agglomerative (Bottom-Up) Clustering

- Most common hierarchical method: **agglomerative clustering**.
 1. Starts with **each point in its own cluster**.
 2. Each step **merges the two “closest” clusters**.
 3. **Stop with one big cluster** that has all points.



[Animation](#)

Agglomerative (Bottom-Up) Clustering

- Reinvented by different fields under different names (“UPGMA”).
- Needs a “distance” between two clusters.
- A standard choice: distance between means of the clusters.
 - Not necessarily the best, many choices exist (bonus slide).
- Cost is $O(n^3d)$ for basic implementation.
 - Each step costs $O(n^2d)$, and each step might only cluster 1 new point.

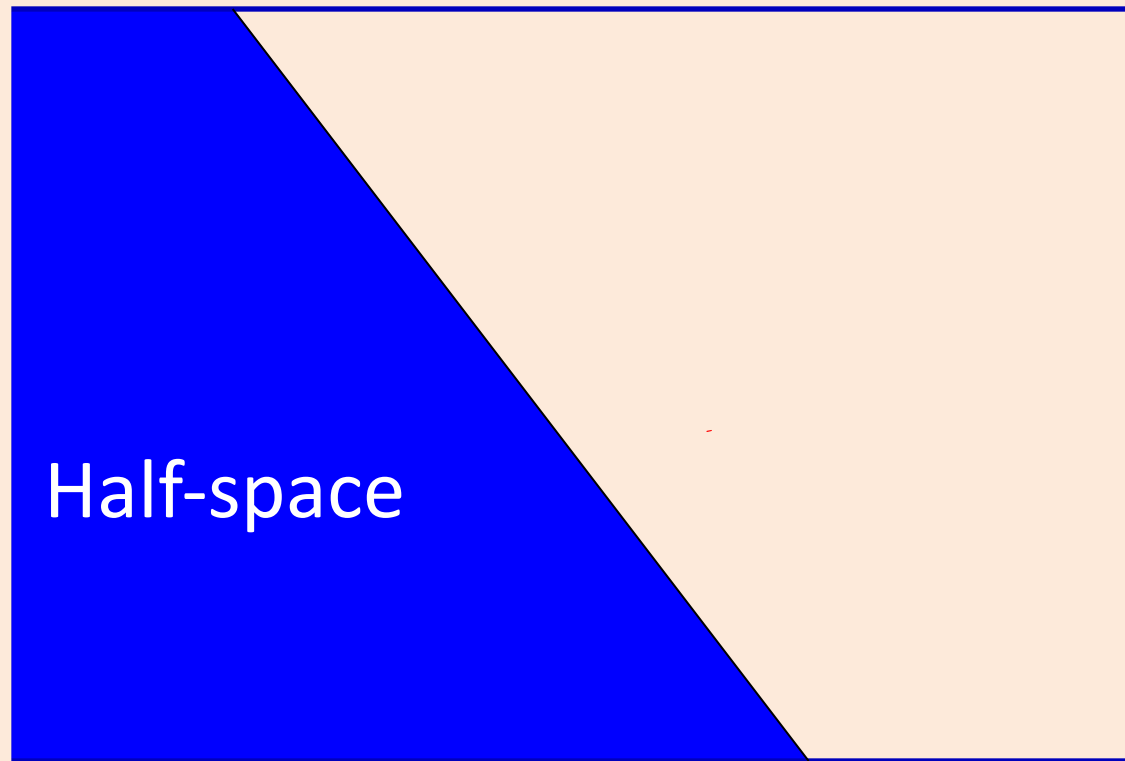
Summary

- **Vector quantization:**
 - Compressing examples by replacing them with the mean of their cluster.
- **Shape of K-means clusters:**
 - Partitions space into convex sets.
- **Density-based clustering:**
 - “Expand” and “merge” dense regions of points to find clusters.
 - Not very sensitive to initialization or outliers.
 - Useful for finding non-convex connected clusters.
- **Ensemble clustering:** combines multiple clusterings.
 - Can work well but need to account for **label switching**.
- **Hierarchical clustering:** more informative than fixed clustering.
- **Agglomerative clustering:** standard hierarchical clustering method.
 - Each point starts as a cluster, sequentially merge clusters.

Why are k-means clusters convex?

- K-means clusters are formed by the **intersection** of **half-spaces**.

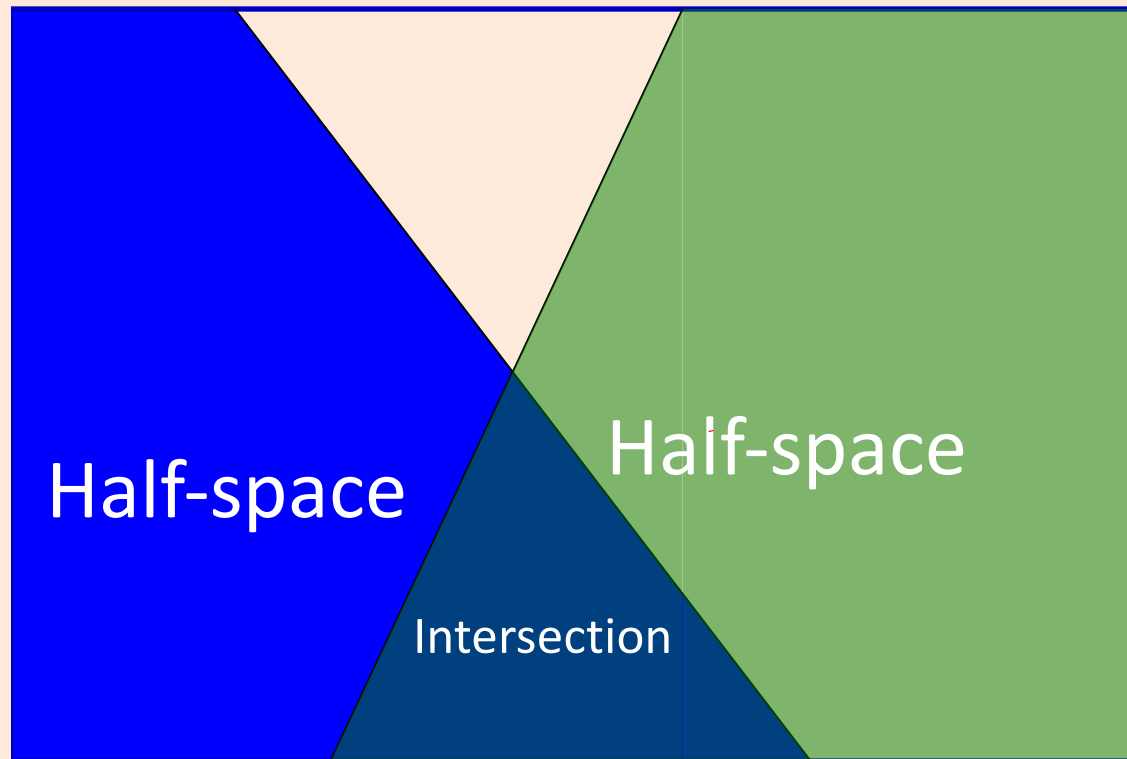
Half-space is Set of points satisfying a linear inequality, like $\sum_{j=1}^d a_j x_j \leq b$



Why are k-means clusters convex?

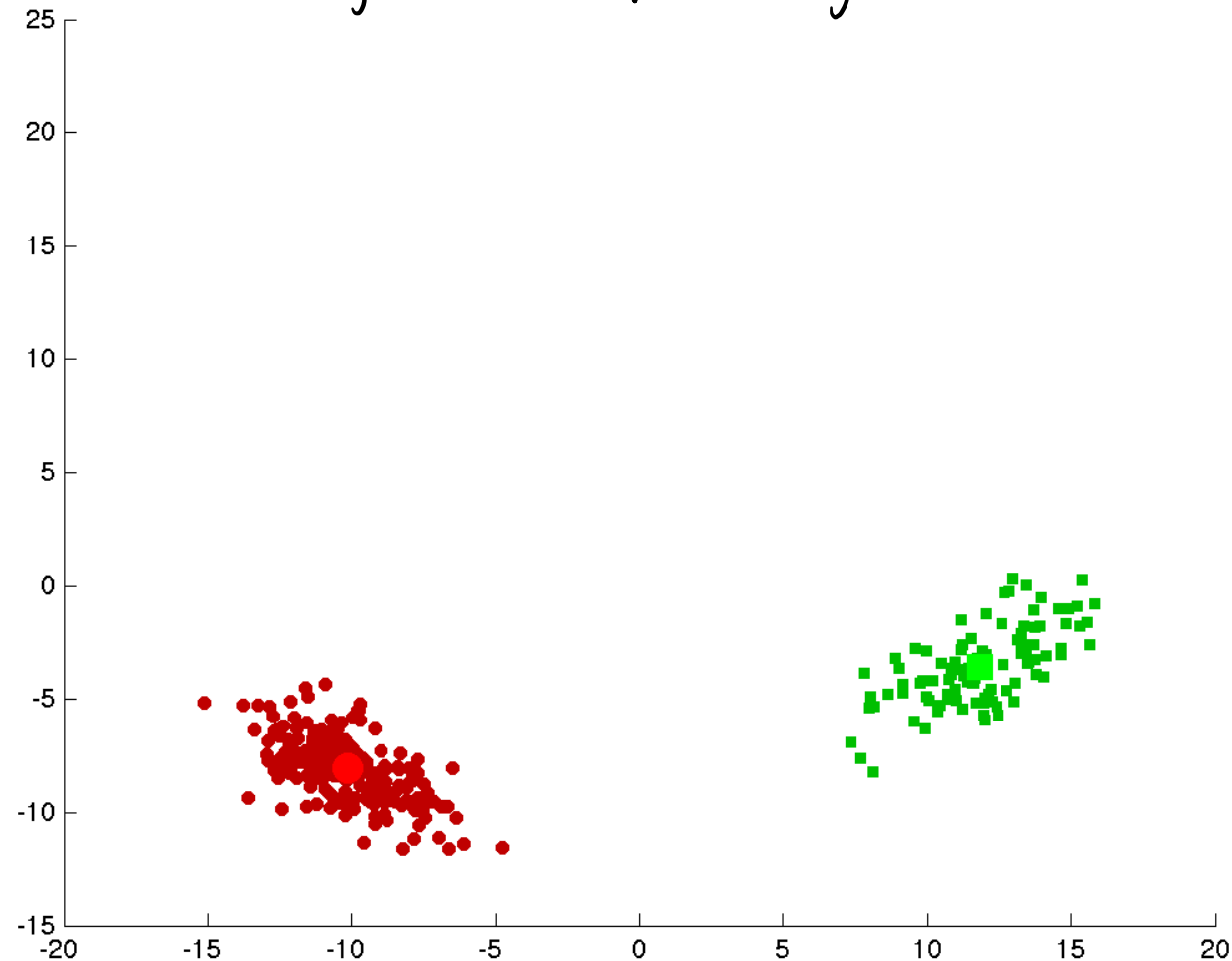
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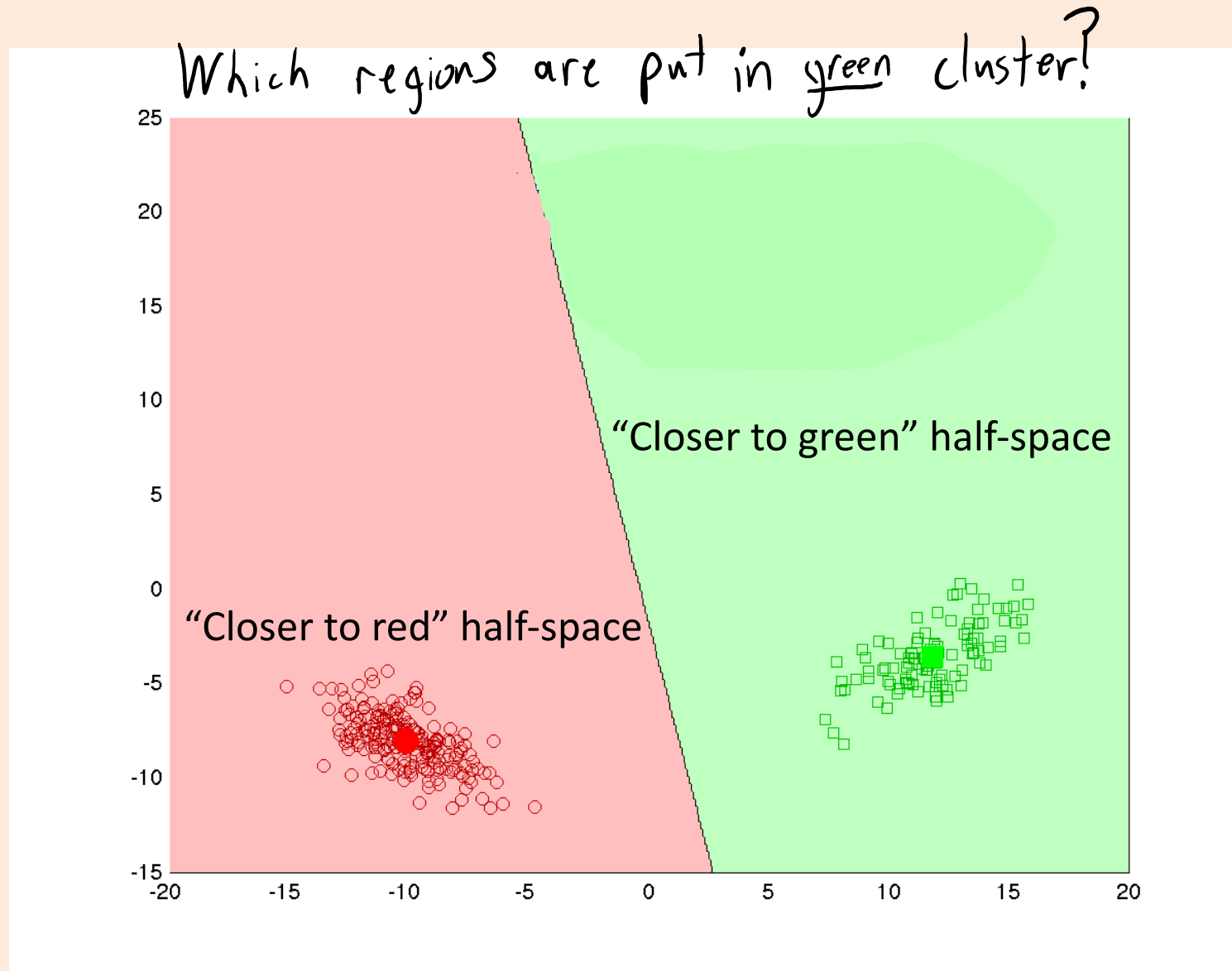


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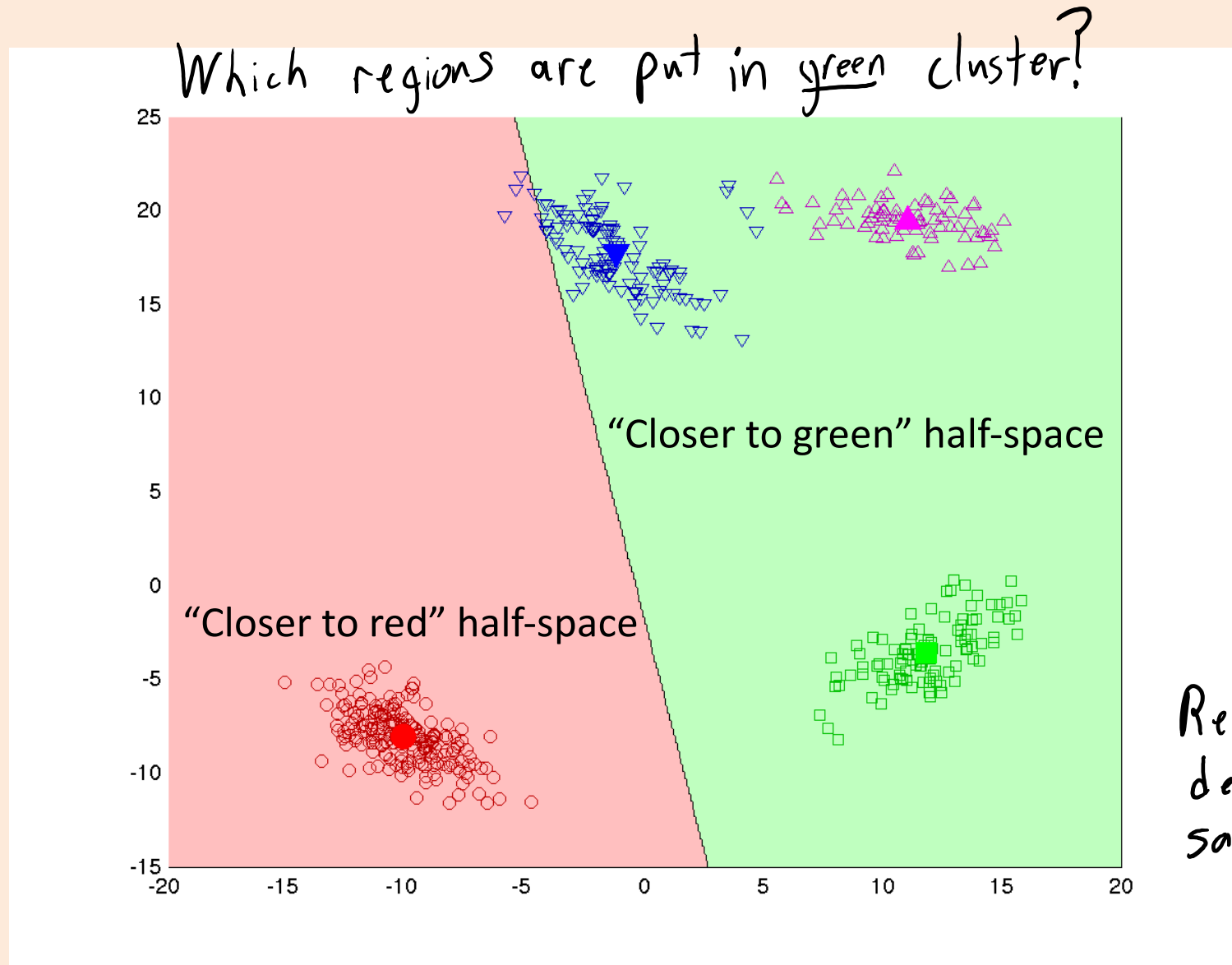
Which regions are put in green cluster?



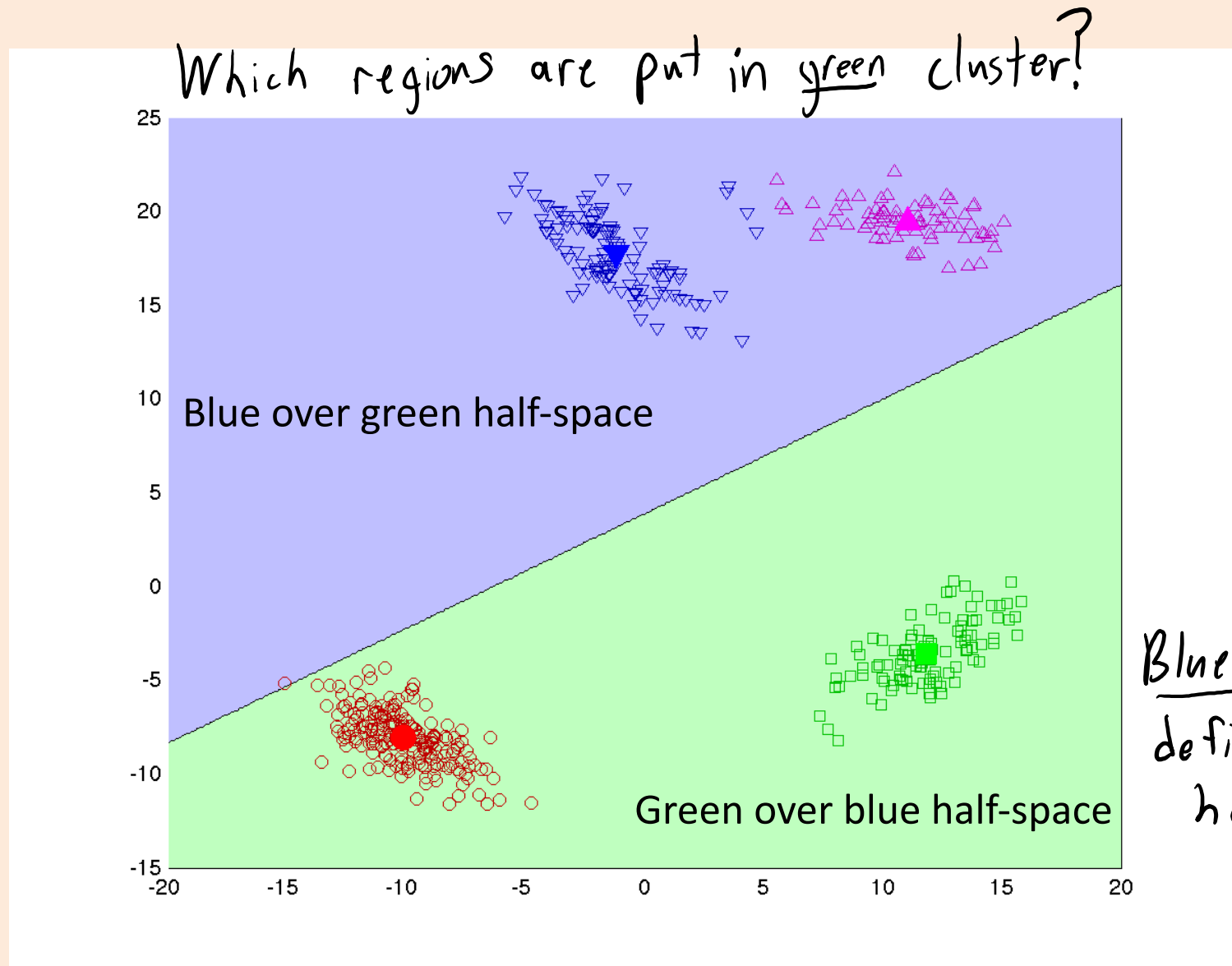
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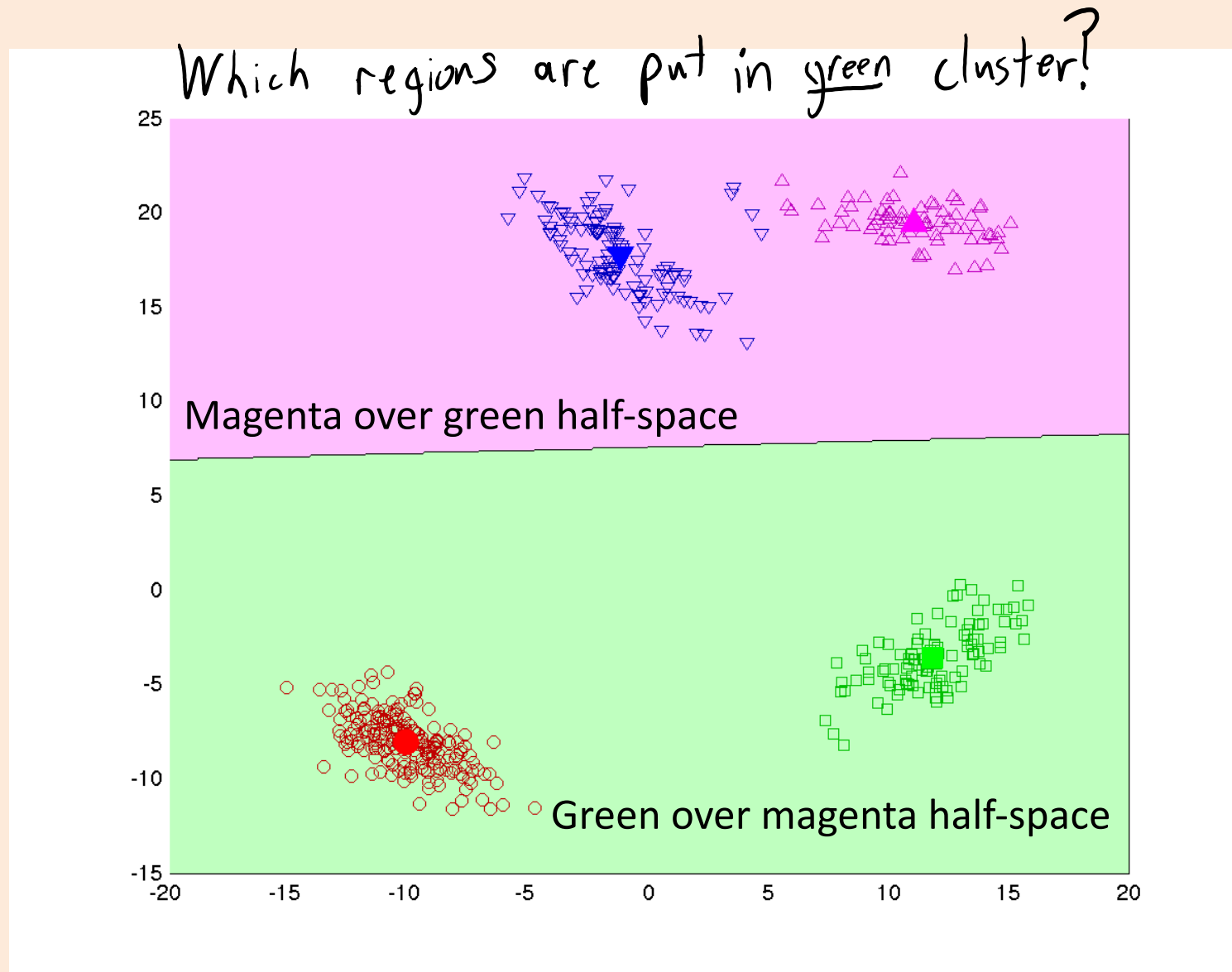


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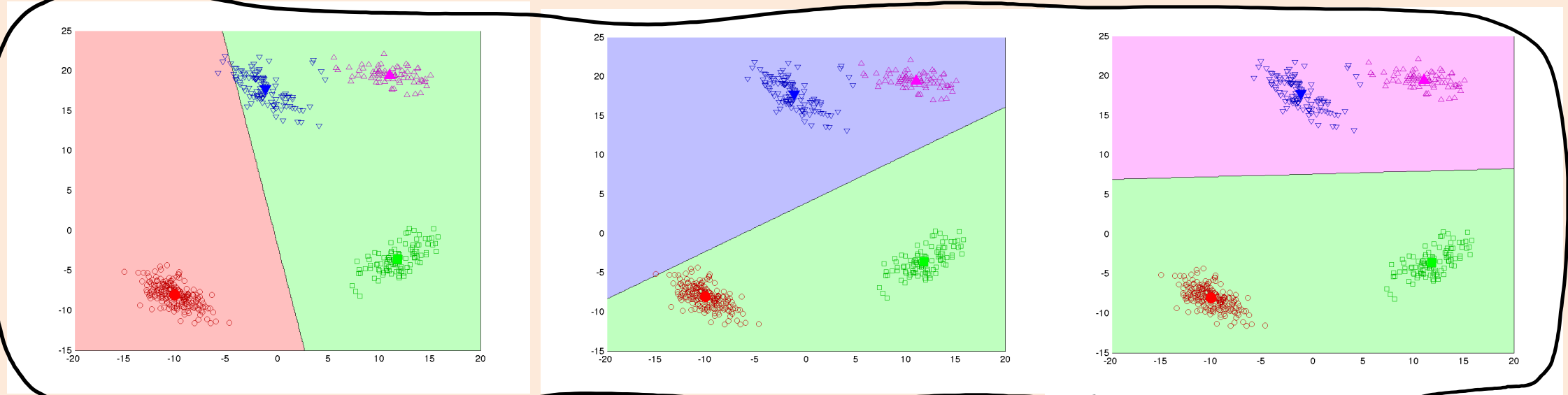


Blue vs. green decision
defines different
half-spaces.

Why are k-means clusters convex?

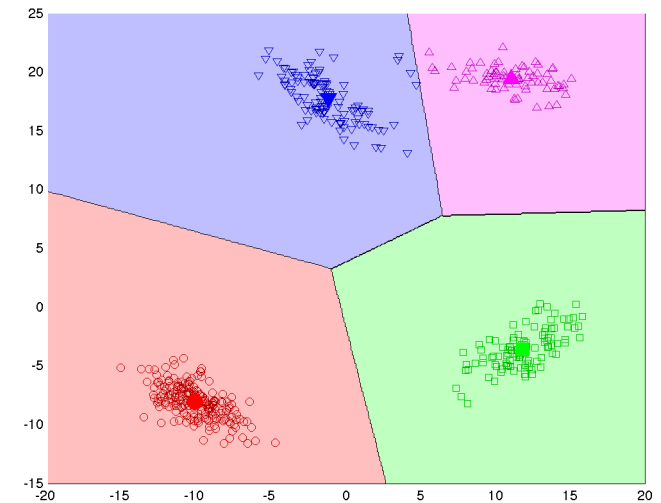


Why are k-means clusters convex?



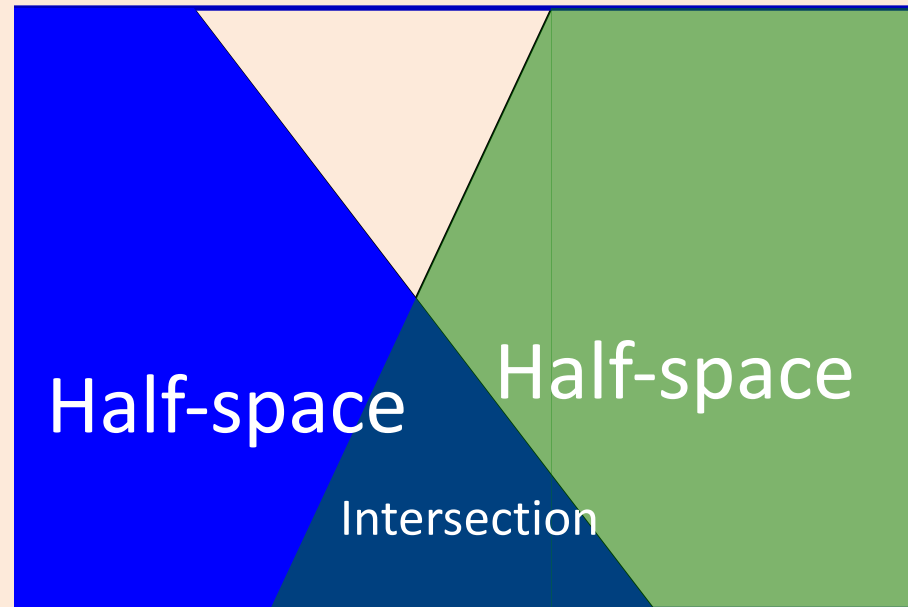
→ Green "cluster" is the intersection of these three half-spaces.

Here is what the four clusters look like:



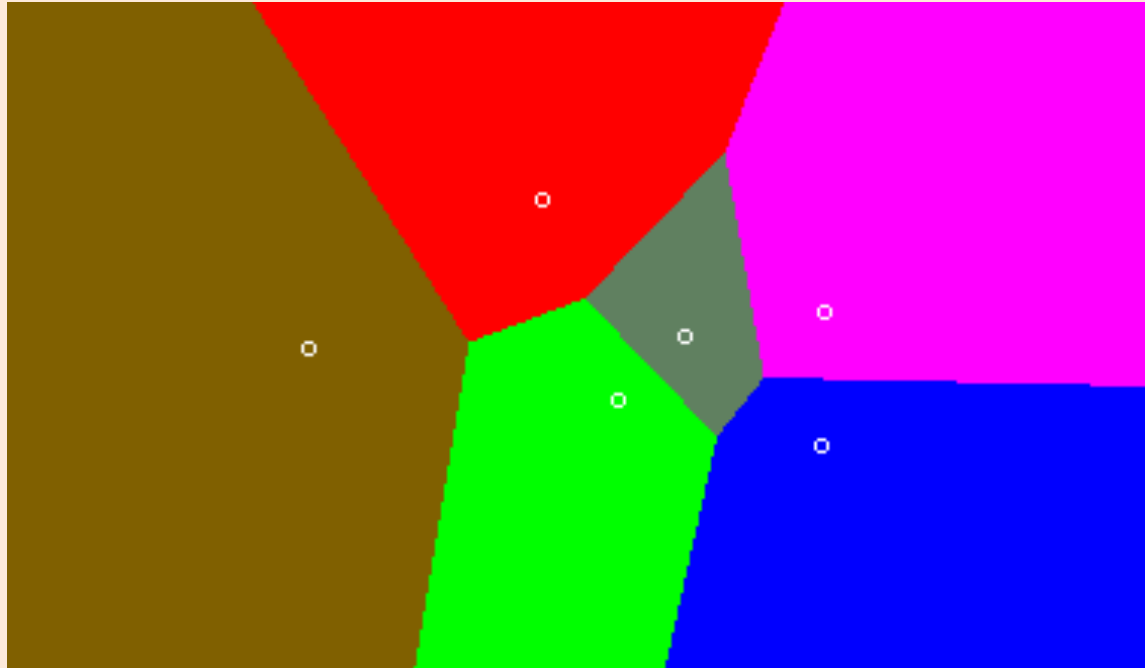
Why are k-means clusters convex?

- Half-spaces are convex sets.
- Intersection of convex sets is a convex set.
 - Line segment between points in each set are still in each set.
- So intersection of half-spaces is convex.



Voronoi Diagrams

- The k-means partition can be visualized as a [Voronoi diagram](#):



- Can be a useful visualization of “nearest available” problems.
 - E.g., [nearest tube station in London](#).

Density-Based Clustering Runtime

? question ★

stop following

72 views

Actions ▼

DBSCAN Training time & Testing time

This is a follow-up inquiry post with Mike about the DBScan, would like to know:

1. Training runtime of DBScan, under k iterations (training set X has n examples and d features)
2. Testing runtime for a single example in DBScan; Testing runtime for test set of size t in DBScan,

i **the instructors' answer,** *where instructors collectively construct a single answer*

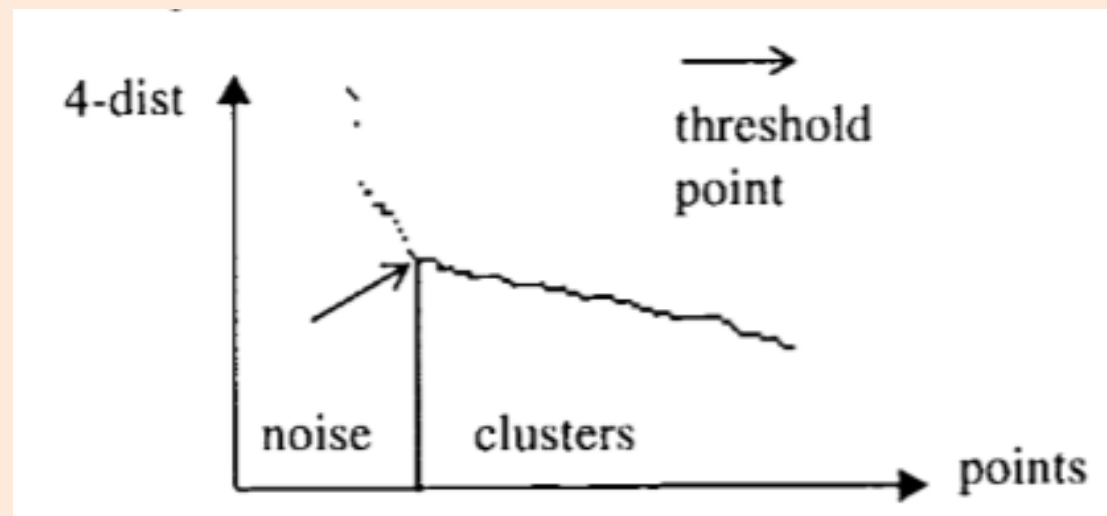
For training, you'll check that each point is a core point exactly once. This check costs $O(nd)$ since you measure the distance to each other point, leading to a total training cost of $O(n^2d)$.

(There are ways to speed this up, like grid-based pruning.)

We didn't define how to apply the DBSCAN model to test data. But a plausible way is to test if the new point is a neighbor of any existing core points. If you have m core points, you would be able to do this in $O(md)$.

“Elbow” Method for Density-Based Clustering

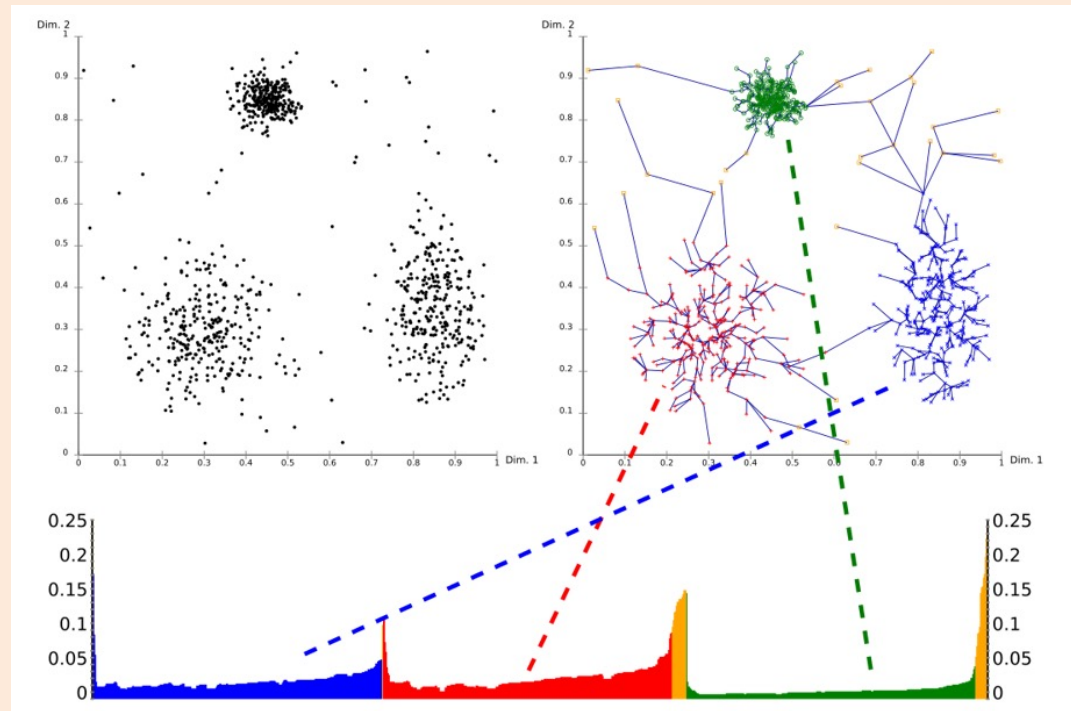
- From the original DBSCAN paper:
 - Choose some ‘k’ (they suggest 4) and set minNeighbours=k.
 - Compute distance of each points to its ‘k’ nearest neighbours.
 - Sort the points based on these distances and plot the distances:



- Look for an “elbow” to choose ϵ .

OPTICS

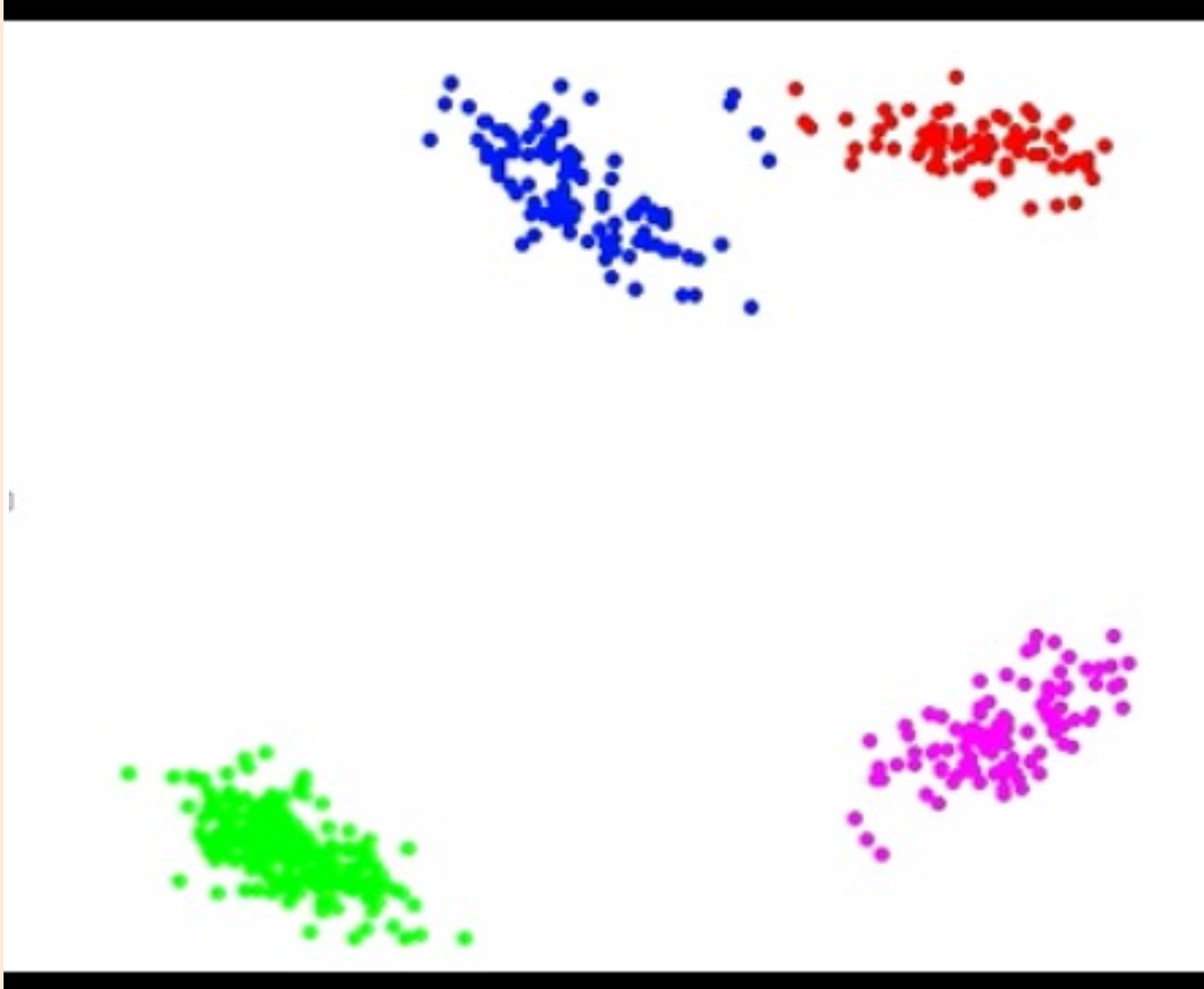
- Related to the DBSCAN “elbow” is “OPTICS”.
 - Sort the points so that neighbours are close to each other in the ordering.
 - Plot the distance from each point to the next point.
 - Clusters should correspond to sequencers with low distance.



UBClustering Algorithm

- Let's define a new ensemble clustering method: **UBClustering**.
 1. Run k-means with 'm' different random initializations.
 2. For each example i and j :
 - Count the number of times x_i and x_j are in the same cluster.
 - Define $p(i,j) = \text{count}(x_i \text{ in same cluster as } x_j)/m$.
 3. Put x_i and x_j in the same cluster if $p(i,j) > 0.5$.
- Like DBSCAN **merge clusters** in step 3 if i or j are already assigned.
 - You can implement this with a DBSCAN code (just changes "distance").
 - Each x_i has an x_j in its cluster with $p(i,j) > 0.5$.
 - Some points are not assigned to any cluster.

UBClustering Algorithm



It looks like DBSCAN, but far-away points will be assigned to a cluster if they always appear in same cluster as other points.

Distances between Clusters

- Other choices of the distance between two clusters:
 - “Single-link”: minimum distance between points in clusters.
 - “Average-link”: average distance between points in clusters.
 - “Complete-link”: maximum distance between points in clusters.
 - Ward’s method: minimize within-cluster variance.
 - “Centroid-link”: distance between a representative point in the cluster.
 - Useful for distance measures on non-Euclidean spaces (like Jaccard similarity).
 - “Centroid” often defined as point in cluster minimizing average distance to other points.

Cost of Agglomerative Clustering

- One step of agglomerative clustering costs $O(n^2d)$:
 - We need to do the $O(d)$ distance calculation between up to $O(n^2)$ points.
 - This is assuming the standard distance functions.
- We do at most $O(n)$ steps:
 - Starting with 'n' clusters and merging 2 clusters on each step, after $O(n)$ steps we'll only have 1 cluster left (though typically it will be much smaller).
- This gives a total cost of $O(n^3d)$.
- This can be reduced to $O(n^2d \log n)$ with a priority queue:
 - Store distances in a sorted order, only update the distances that change.
- For single- and complete-linkage, you can get it down to $O(n^2d)$.
 - “SLINK” and “CLINK” algorithms.

Bonus Slide: Divisive (Top-Down) Clustering

- Start with all examples in one cluster, then start dividing.
- E.g., run k-means on a cluster, then run again on resulting clusters.
 - A clustering analogue of decision tree learning.

